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Whole genome resequencing of *Heliconius* butterflies revolutionizes our view of the level of admixture between species

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Abstract

The theory of "genomic islands of speciation" has been extensively debated during the last decade. This thesis not only supports this theory, but provides evidence that challenges previous beliefs on the level of admixture between species. The recently published *Heliconius melpomene* genome project reported apparent genomic paraphyly of *H. pardalinus* with regard to *H. elevatus* (Heliconius Genome Consortium 2012). Here, we investigate this pair of butterfly species more fully, firstly by using whole-genome resequence data, and secondly by analyzing additional geographic populations of both species, as well as outgroup taxa. Using a nuclear whole-genome phylogenetic analysis we also confirm that *H. elevatus* is paraphyletic. The genome-wide phylogenetic signal in *H. pardalinus* and *H. elevatus* does not indicate expected mutual monophyly of each species as it seems strongly distorted by a high level of admixture. However, several regions of the genome remain differentiated and do show the presumably original phylogenetic signal with mutual monophyly of *H. pardalinus* and *H. elevatus*. The genomic background is so homogenized that its level of differentiation ($F_{ST} \sim 0.03$) virtually implies panmixia. The pattern of a high level of homogenization across the genome with several regions of differentiation was consistent with a number of other statistics such as absolute divergence D_{xy} , nucleotide polymorphism π , number of fixed differences and with a sliding window phylogeny. The identified genomic islands of divergence comprise genes responsible for wing-patterning and chemosensation in *Heliconius* and some of these genes are found to be under positive selection, suggesting possible candidates of speciation.

Introduction

In the emerging field of speciation genomics, genetic patterns of divergence in sympatry, with no geographic barriers to gene flow between species, is a topic of great interest. The genome-wide analysis of mosquito *Anopheles gambiae* M and S forms (Turner *et al.* 2005) for the first time empirically demonstrated that genetic differentiation between nascent species can be limited to a few "genomic islands of speciation", while the background genome remains highly homogenized. Several subsequent studies on other species (Hohenlohe *et al.* 2010, 2012; Nadeau *et al.* 2012; Ellegren *et al.* 2012; Martin *et al.* 2013) also demonstrated that interspecific differentiation could be driven by a few divergent loci with a considerable admixture throughout the rest of the genome. These findings support the genic view of speciation (Wu 2001), in that complete whole genome reproductive isolation is not essential for species emergence.

However, more recent investigations of *A. gambiae* have cast doubts on the probability of high levels of gene flow between M and S forms. The "islands of speciation" may be artifacts of recent selective sweeps rather than evidence for speciation with gene flow (Turner & Hahn 2010; White *et al.*

2010; Hahn *et al.* 2012). In the light of this skepticism, this thesis aims to investigate patterns of divergence in new whole-genome sequence data from neotropical butterfly species of the genus *Heliconius*, one of the groups in which speciation-with-gene-flow is highly probable (Salazar *et al.* 2010; Nadeau *et al.* 2012; Heliconius Genome Consortium 2012; Martin *et al.* 2013).

Heliconius butterflies have bright and very diverse wing color patterns. These color patterns act as warning colors to predators, whilst simultaneously functioning as one of the cues in assortative mating. Many of the species are involved in Müllerian mimicry, which is mimicry among reciprocally distasteful species. It has also been shown that Müllerian mimicry can play a role in wing color divergence and ecological speciation (Jiggins *et al.* 2001; Chamberlain *et al.* 2009). In sympatry, species differences are usually maintained by behavioral isolation (Jiggins *et al.* 2004; Merrill, Van Schooten, *et al.* 2011; Merrill, Gompert, *et al.* 2011, but see Naisbit *et al.* 2001), which sometimes fails whereupon different species occasionally hybridize (Mallet *et al.* 2007; Dasmahapatra *et al.* 2007). Although the hybridization rate is very low (1/1000 mates per generation) (Mallet *et al.* 2007), over long periods it may substantially inhibit genomic divergence between species. Interspecific hybridization may also result in introgression of ecologically adaptive genes and facilitate speciation. There are several *Heliconius* species (*H. heurippa*, *H. timareta*, *H. elevatus*) that have been shown to have introgressions of the color pattern region from one species into the background of another (Salazar *et al.* 2010; Heliconius Genome Consortium 2012).

Evidence of hybridization in many sympatric *Heliconius* species and races which, unlike *A. gambiae*, are clearly distinctive phenotypically, make these butterflies a good model to test the hypothesis of "genomic islands of speciation". However, clear examples of "islands of divergence" in *Heliconius* were found only in Peruvian and Panamanian color-pattern races of *H. melpomene* (Nadeau *et al.* 2012; Martin *et al.* 2013). These races hybridize freely in the center of their hybrid zone and do not show strong isolation mechanisms. The "islands of divergence" are mainly attributed to the loci controlling wing color patterning and their divergence is rather driven by locally adapted mimicry than reproductive isolation in the traditional sense. A broad scale genomic study of recognized species that locally hybridize, such as *H. timareta*, *H. cydno* and *H. melpomene*, demonstrates that these species are at a stage with more genomically widespread divergence when "islands of divergence" are not very noticeable (Martin *et al.* 2013). Thus, islands of divergence in *Heliconius* so far seem to have been found only between geographic races within a species, and there is a lack of evidence for "genomic islands" in pairs of sympatric taxa that coexist in widespread sympatry with speciation process.

In this study, we focus on two species *H. elevatus* and *H. pardalinus*, which have diverged recently, probably as a result of the *H. pardalinus*-like ancestor of *H. elevatus* acquiring a rayed mimetic color pattern from *H. melpomene* (Heliconius Genome Consortium 2012). These species are distributed

across the Amazon basin and some adjacent regions, where they are broadly sympatric (Fig. 1). The range of *H. elevatus* extends more to the north, while the distribution of *H. pardalinus* covers some regions at the edge of Amazon, where *H. elevatus* does not occur. *H. elevatus* and *H. pardalinus* have very distinct wing color patterns (Fig. 2, 3), mimic totally different species (*H. elevatus* mimics other rayed *Heliconius*, such as *H. erato* and *H. melpomene*; *H. pardalinus* mimics *Melinaea* and other "tiger-pattern" ithomiines and heliconiines) and show assortative mating in experiments (Dasmahapatra & Rosser, in prep.). Natural interspecific hybrids are known but very rare. Most of the previous phylogenetic studies based on a few markers found that *H. elevatus* and *H. pardalinus* were sister species (Beltrán *et al.* 2007; Dasmahapatra *et al.* 2007; Mallet 2009). A more recent phylogenetic study based on RAD sequencing (Heliconius Genome Consortium 2012) suggested that *H. pardalinus* is a paraphyletic species because Amazonian *H. elevatus* groups together with Amazonian *H. pardalinus butleri*, while allopatric *H. pardalinus sergestus* from the Mayo/Huallaga valley system of Peru, where *H. elevatus* is unknown, is strongly divergent from the Amazonian forms of both species. Two extreme possibilities could explain these results:

(1) The original morphology-based species-level taxonomy of this group is profoundly incorrect, so that, Amazonian *H. elevatus*, Amazonian *H. pardalinus* and *H. p. sergestus* should all be viewed as separate species in their own rights.

(2) The original morphology-based level taxonomy represents the true pattern of splitting of the species, but ongoing gene flow between the species in the Amazon has contributed to genome-wide admixture in the Amazon basin that has led to the genome-wide phylogenetic analyses giving incorrect estimates of history. If so, the Amazonian forms would be distinct at least at parts of the genome that remain differentiated due to association with the maintenance of species differences.

We therefore set about testing these possibilities. Considering that the inconsistency of the whole genome phylogeny with the species phylogeny could be due to admixture, we assume that the level of genomic admixture between sympatric *H. pardalinus* and *H. elevatus* resulting from occasional hybridization exceeds 50%, a hitherto hardly imaginable fraction of the genome.

Material and Methods

Samples

For each species in every geographic population used in this study we sequenced whole genomes of four individuals (2 males and 2 females). The sympatric populations were represented by *H. elevatus pseudocupidineus*, *H. pardalinus* ssp. nov. and *H. pardalinus butleri* from Peru, and *H. pardalinus orteguaza* and *H. elevatus* from Ecuador. *H. elevatus bari* from French Guiana was used as an allopatric population of *H. elevatus*, *H. pardalinus sergestus* served the same role for *H. pardalinus*,

since Peruvian *H. elevatus* are known from sites very close to but not sympatric with *H. p. sergestus*. *H. hecale felix* from Peru represented a closely related outgroup for phylogenetic analysis, while *H. melpomene aglaope*, *H. melpomene amaryllis* and *H. timareta ssp. nov.* acted as a more distant outgroup. Localities from which specimens were collected are shown on Fig. 1.

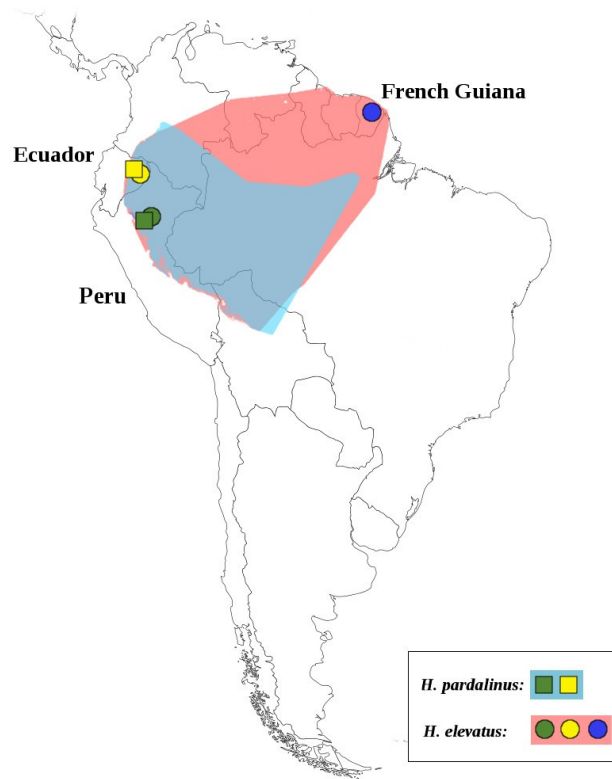


Figure 1. The distribution ranges of *H. elevatus* and *H. pardalinus* and the collection points used in this study. Species ranges are shown according to Rosser et al. (2012). Color of the points of collection represents the country, while shape symbolizes species belonging. Credits for the original blank map: [Roke](#).

Sequencing and genotype calling

The genomes of all the specimens were sequenced using paired-end HiSeq Illumina 100 bp sequencing technology. The reads were aligned to scaffolds of the *H. melpomene* reference genome (version 1.2) (Heliconius Genome Consortium 2012; Martin *et al.* 2013) using Stampy v1.0.20 (Lunter & Goodson 2011). PCR duplicates were removed using Picard Tools 1.88 (<http://picard.sourceforge.net>). Indels realignment, depth of coverage estimation and genotype calls were performed in Genome Analysis Tool Kit v2.4.9 (GATK) (DePristo *et al.* 2011). Genotypes were used only with minimum SNP quality of 30, minimum genotype quality of 30, minimum mapping quality of 30 and coverage from 10 reads to 300 reads. The upper coverage threshold was set up to avoid false calls due to mis-mapping of reads from highly repetitive regions. Z chromosomal scaffolds were reassigned to the Z chromosome as in Martin *et al.* (2013). This reasonably increased the length of the Z chromosome to ~3x the size of the original assembly (Heliconius Genome Consortium 2012).

MtDNA was separated from the whole-genome sequences as one scaffold and analyzed separately. Genotype calling for mtDNA was performed with coverage of 10-500 reads to account for gene copy number in the original sample.

Phylogenetic analysis

A nuclear genome phylogeny of all 35 individuals included in this study was reconstructed by neighbour-joining (NJ) phylogenetic analysis using the alignment of only complete and polymorphic sites (66Mb). Monomorphic and incomplete sites were removed to reduce the data set size as the phylogenetic programs available to us (PhyML v. 20120412, RaxML 7.4.2, FastTree v. 2.1.7) were unable to handle the entire dataset due to RAM issues. Mitochondrial genome phylogeny was reconstructed separately for the same 35 individuals, using both NJ in PhyML v. 20120412 (Guindon *et al.* 2010) and maximum likelihood (ML) methods with the GTRGAMMA model in RaxML 7.4.2 (Stamatakis *et al.* 2012), with 1000 bootstrap replicates.

To explore the phylogenetic signal across the genome, we generated a ML phylogenetic tree based on the GTRGAMMA model in RaxML 7.4.2 (Stamatakis *et al.* 2012) for every non-overlapping 50kb window, filtered for at least 20kb of non-missing sites in the alignment of each window. Both monomorphic sites and sites with some (but not all) missing data were retained for these phylogenetic analyses. The analysis of frequencies of different topologies in the obtained set of trees was performed using Newick Utilities v. 1.6 (Junier & Zdobnov 2010).

We also reconstructed ML and NJ trees from the alignment generated by concatenating all sites within the highest F_{ST} peaks > 0.4 . Bootstrap support of these two trees was based on 100 replicates. The same sites from F_{ST} peaks were also concatenated within each chromosome and NJ trees for differentiation sites of every chromosome were obtained.

Population genetic analysis

We calculated nucleotide diversity π and Tajima's D within each species, and level of differentiation F_{ST} , absolute divergence D_{xy} , number of fixed differences between species using the EggLib Python module (De Mita & Siol 2012). The calculation was performed in sliding windows of 50 kb moving in 25kb intervals across the genome. Windows with less than 5000 complete sites were rejected.

In addition, to assess the level of admixture across the genome we applied an f4-population test, which was recently introduced to test for admixture in human populations (Reich *et al.* 2009, 2012). This test fits a phylogenetic tree to the measured genetic drift. The assumption is that if populations are

not admixed, there is no correlation in genetic drift (allele frequencies) between sympatric non-sister taxa. Given the populations Y, Z, W, X with the unrooted topology ((Y, Z), (W, X)), we calculate D statistics inferred from the f4 test using formula as in Patterson et al. (2012):

$$D = \frac{\sum_i Num_i}{\sum_i Den_i}$$

$$Num_i = (w - x)(y - z)$$

$$Den_i = (w + x - 2wx)(y + z - 2yz)$$
(1)

D -statistics were estimated for each informative site (i.e. biallelic SNPs with at least two data points available per site per group) and averaged for every 50 kb window across the genome with moving intervals of 25kb and a window rejection criterion of <5000 informative sites. The D ranges from -1 to 1 where a value of 0 indicates that neither Y nor Z are correlated with W and X, thus there is no admixture. A positive deviation from 0 is indicative of admixture between Y and W or between Z and X, or both, while a negative D implies admixture between Y and X, or Z and W, or both.

Means of π , *Tajima's D*, D_{xy} and number of fixed differences in F_{ST} peaks ($F_{ST} > 0.4$) were contrasted to means of the same statistics in the background genome ($F_{ST} < 0.4$). In this analysis we used non-overlapping 50kb genomic windows. The 97.5 and 2.5 percentile confidence intervals and p-values were computed by bootstrap resampling of 10000 times in R (R Core Team 2012). The Z chromosome was analyzed separately by its comparison versus the rest of the genome. Besides π , *Tajima's D*, D_{xy} and number of fixed differences, we also compared mean F_{ST} of the Z chromosome and the rest of the genome.

Analysis of the gene annotation and gene ontology

To infer the functional role of the divergent regions ($F_{ST} > 0.4$), we investigated which genes stand behind the discovered F_{ST} peaks and whether they are enriched for any gene ontology (GO) terms, terms that allow to infer functions of the genes (Gene Ontology Consortium 2000, 2008). In these analysis we used the *H. melpomene* reference genome annotation v1.1 (Heliconius Genome Consortium 2012). Functional characterization of the F_{ST} peaks was done by retrieving the gene annotation from the downloaded gff3 file using custom scripts. A GO term enrichment analysis was carried out in g:Profiler (Reimand et al. 2011), which on the date of the analysis (19 Nov 2013) was based on the *H. melpomene* reference genome annotation v1.1.

dN/dS calculation

We used nucleotide variation in synonymous (*dS*) and non-synonymous (*dN*) sites to identify signs of selection from *dN/dS* ratio (ω). In the absence of selective pressure, the probability of synonymous and non-synonymous substitutions are equal ($\omega=1$). If a gene is conserved, constrained non-synonymous variations that replace amino acids in a protein are removed by selection making ω smaller than 1. Under the positive selection, change of amino acids is favored and ω becomes greater than 1. To estimate ω , we implemented the branch model and the branch-site model within the PAML 4.7 package in the program *codeml* (Yang 2007). This analysis was based on the “true” species phylogenetic tree ((((*H. p. sergestus*, Amazonian *H. pardalinus*) *H. hecale*), (*H. e bari*, Amazonian *H. elevatus*)), *H. melpomene* outgroup).

In the branch model, we tested the following hypotheses :

H0: ω does not vary across branches (model=0)

H1: ω has two different ratios: one for *H. pardalinus* and one for other branches (model=2).

H2: ω has two different ratios: one for *H. elevatus* and one for other branches (model=2)

H3: ω has three different ratios: one for *H. elevatus*, one for *H. pardalinus* and one for other branches in the phylogeny (model=2).

H0 vs. H1 and H0 vs. H2 were compared with d.f.=1 and H0 vs. H3 with d.f.=2 by the likelihood ratio test (LRT).

The branch-site model allows ω to vary across sites in the sequence and across branches the phylogeny simultaneously. In this model, a tested phylogenetic branch is named a foreground branch and it is compared against other branches that are named background branches. The following hypotheses were tested using either *H. elevatus* or *H. pardalinus* as foreground branches:

H0: ω is fixed to 1 across the phylogeny (model = 2, fix_omega = 1, NSsites = 2)

H1: ω is estimated separately for the foreground and background branches (model = 2, fix_omega = 0, NSsites = 2).

H0 vs. H1 were compared with d.f.=1 by the LRT.

We analyzed several selected genes: odorant binding protein *HmOBP20* (HMEL008689), olfactory receptor *HmOR30* (HMEL006027), *fizzy*-like (HMEL000025), *optix* (HMEL001028), *WntA* (HMEL018100), *penguin*-like (HMEL000041) genes from the regions of divergence and odorant binding protein genes *HmOBP29* (HMEL006095), *HmOBP31* (HMEL007613), *HmOR36* (HMEL017068) from the genomic background. The coding sequences were reconstructed by concatenating coding DNA sequence (CDs) for each gene. The alignment of each gene was verified by translating nucleotide sequences into amino acid sequences and checking it by eye. Heterozygous sites were retained and treated as ambiguous characters by PAML (cleandata = 0). All genes were the same

length as in the reference genome, except the *fizzy*-like gene which had a gap of missing data (between the positions 31 and 154) that was removed.

Results

Sequencing and genotype calling

Whole genome paired-end Illumina sequencing of 35 individuals gave coverage per individual of x19-44 (average x32). Reads aligned to the *H. melpomene* reference genome (Heliconius Genome Consortium 2012) produced 242 million sites (90% of the reference genome) with information available for at least one individual, 66 million complete sites (without missing data) (25% of the reference genome), and 13,6 million complete polymorphic sites (5% of the reference genome).

Alignment of mitochondrial genomes resulted in 15kb with information available for at least one individual and it covered 99% of the *H. melpomene* mitochondrial reference genome. Removing all incomplete sites left us 12kb (78% of the reference genome). Polymorphic sites were represented by 1.1 kb.

Whole genome phylogeny

The nuclear phylogenetic tree was reconstructed using 66Mb sequence alignment of 35 individuals. Previously paraphyly of *H. pardalinus* only was reported (Heliconius Genome Consortium 2012), while our new phylogenetic tree shows that *H. elevatus* is not a monophyletic group either (Fig. 2A). In concordance with the previous results, all Amazonian races of *H. pardalinus* and *H. elevatus* cluster together leaving Peruvian *H. p. sergestus* as a sister group, while *H. e. bari* from French Guiana unexpectedly joins all of them basally. This result puts into question the validity of the taxonomy of *H. elevatus* and *H. pardalinus*.

The phylogenetic tree of the mitochondrial genome was calculated from 15kb alignment. Two mitochondrial phylogenetic lineages within the Amazonian *H. pardalinus* and *H. elevatus* group were discovered (Fig. 2B). One of them is common for most of the Amazonian races of *H. pardalinus* and *H. elevatus*, while another one, a sister group of *H. e. bari* from French Guiana, occurs only in Ecuadorian *H. p. orteguaza* and *H. elevatus*. Surprisingly, the mitochondrial genome of *H. p. sergestus* is phylogenetically closer to *H. hecale* than to any other races of *H. pardalinus*. Thus, the topologies of mitochondrial and nuclear trees neither support mutual monophyly of *H. elevatus* and *H. pardalinus*, nor are consistent with each other.

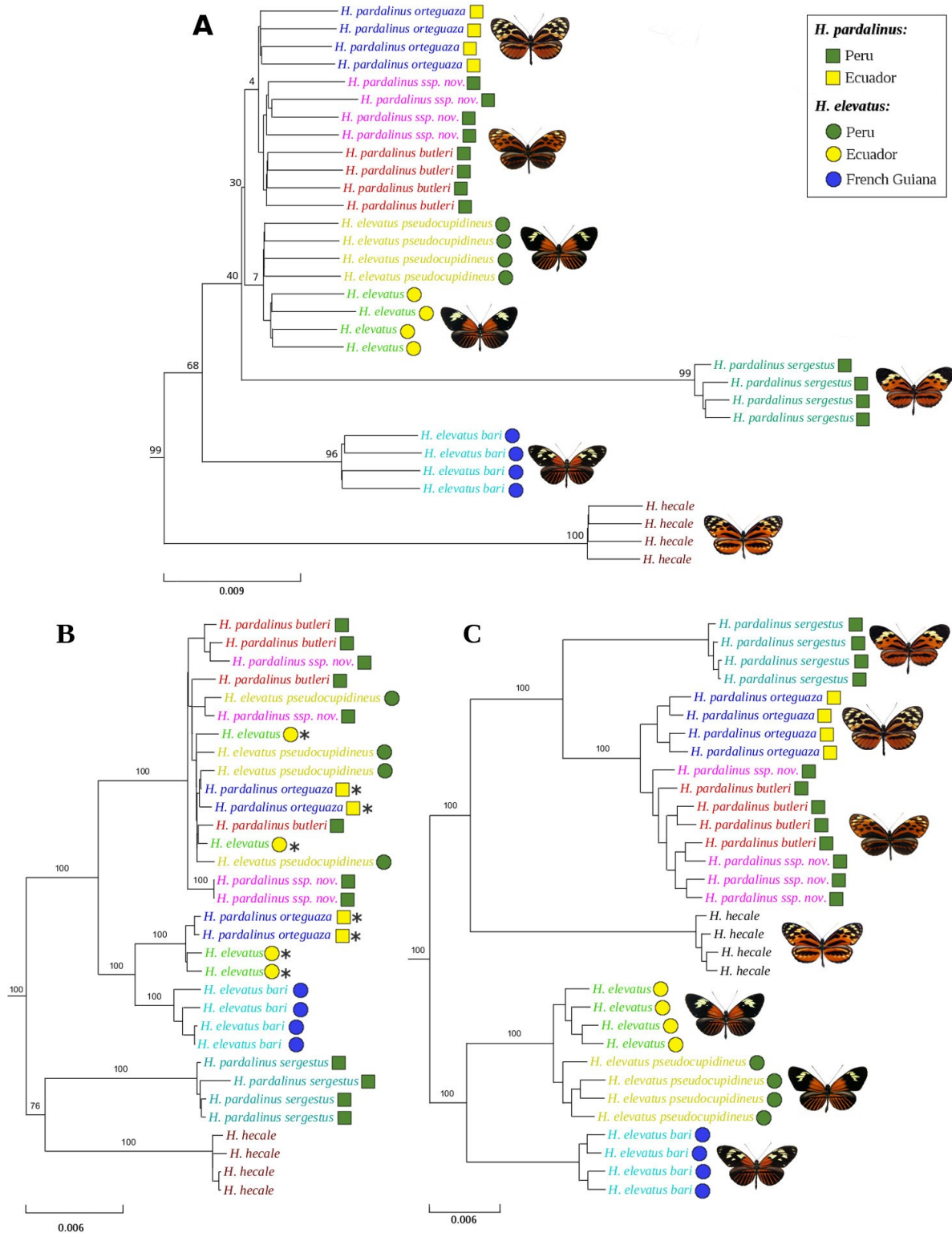


Figure 2. Phylogenetic relationships of *H. elevatus* and *H. pardalinus*. The outgroup (not shown) is represented by *H. m. aglaope*, *H. m. amaryllis* and *H. timareta* ssp. nov. in all trees. **A. The genome-wide tree:** NJ phylogenetic tree of the whole nuclear genome. The node support is provided as the frequency (in percents) of a given phylogenetic group in 5353 ML phylogenetic trees calculated in 50 kb sliding window across the genome. Values equal or smaller than 1% are not shown. **B. The mtDNA tree:** ML phylogenetic trees of the whole mitochondrial genome. Stars distinguish species that comprise two mitochondrial lineages. Bootstrap support of > 50% is shown on the nodes. **D. The “species tree”:** ML phylogenetic tree of concatenated regions of divergence ($F_{ST} > 0.4$) between *H. elevatus* and *H. pardalinus*. Bootstrap support within races is not shown. Credits for the butterflies pictures: [Massimo Demaiò](#).

Besides the discrepancies between the phylogenetic signals of nuclear and mitochondrial genomes, phylogenetic discordance may be also found in different regions of the genome (Garrigan *et al.* 2012; Martin *et al.* 2013). Therefore, we scanned for phylogenetic signal across the genome using 50kb sliding window approach and visualized the results by plotting all the trees (Fig. 3). The density plot of 5353 trees demonstrates that the phylogenetic relationships between Amazonian races of *H. elevatus* and *H. pardalinus* are very blurred across the genome. The frequency of the whole genome topology in these sliding window trees is provided as branch support on Fig. 2A and indicates that the whole genome phylogeny is an average over many discordant trees.

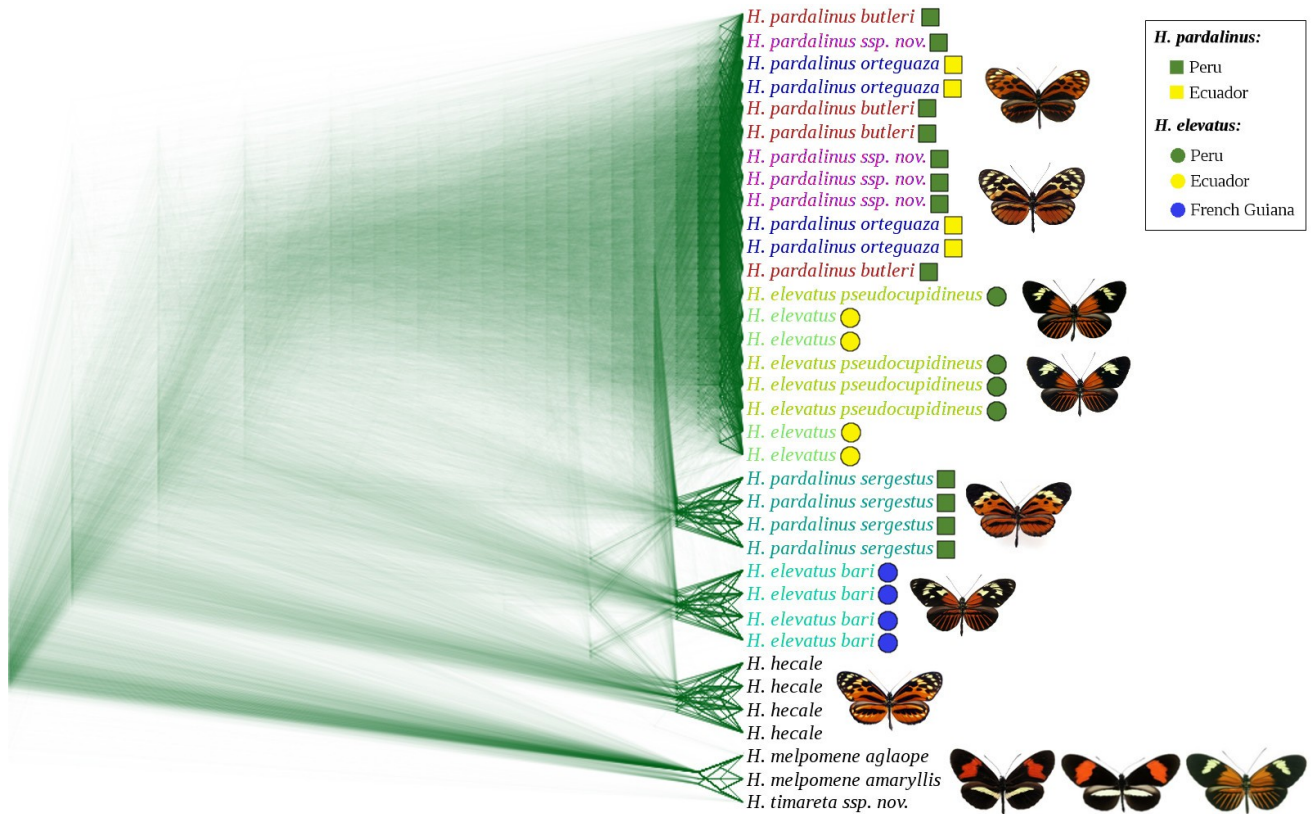


Figure 3. Overlapped ML phylogenetic trees reconstructed from 50kb sliding windows across the nuclear genome. Yellow and green squares next to the species names mark Amazonian *H. elevatus* and *H. pardalinus* collected from Ecuador and Peru respectively. Purple and gray squares mark *H. p. sergestus* and *H. e. bari*, species that are allopatric to Amazonian group. The trees were plotted in DensiTree (Bouckaert 2010). Credits: butterflies pictures by [Michel Cast](#) (*H. m. aglaope* and *H. m. amaryllis*) and [Massimo Demai](#) (others).

Genetic differentiation across the genome

We calculated population differentiation F_{ST} , divergence D_{xy} , nucleotide diversity π , and number of fixed differences between Amazonian races of *H. pardalinus* (*H. p. ssp. nov.*, *H. p. butleri* and *H. p. orteguaza*) and *H. elevatus* (*H. e. pseudocupidineus* and *H. elevatus*) in 50kb windows across the genome. The obtained F_{ST} values are mostly close to 0 (average ~ 0.03) across the whole genome

(Fig. 4) with several regions exceeding the value of 0.4 (average ~ 0.52). The highest peaks reach 0.75 and are situated in or very close to three known color pattern loci *Ac*, *B/D* and *N/Yb/P* on chromosomes 10, 15, and 18, respectively (Joron *et al.* 2006, 2011; Reed *et al.* 2011; Martin *et al.* 2012; Nadeau *et al.* 2012). The F_{ST} peaks (hereafter those > 0.4 but excluding the Z chromosome) comprise 1.4 % (3376690 bp) of the whole genome (242581339 bp). The Z chromosome shows considerable differences in its F_{ST} pattern compared to the autosomes. Unlike most of the genome where F_{ST} forms a flat line with rare spike peaks, the F_{ST} pattern in this sex chromosome is uneven with many narrow small peaks. The mean F_{ST} in the Z was significantly lower than in the autosomes.

The distribution of D_{xy} across the genome (Fig. S5) shows little correspondence with the distribution of F_{ST} . The only noticeable pattern is that D_{xy} decreases towards the edges of chromosomes and is relatively low across the whole Z chromosome. This is similar to the pattern of π , which also decreases towards the edges of autosomes and is very low in the Z chromosome. However, the reduction in π are often found in the regions of F_{ST} peaks as well. The clearest concordance with F_{ST} is observed in the distribution pattern of the number of fixed differences (Fig. S6).

To provide further comparison between conformity and discrepancy in different parameters, we compared their means in the regions of F_{ST} peaks versus genomic background (Fig. 5). Considering that the highest F_{ST} peaks are linked to three color pattern loci that were shown to play major role in the divergence of some *Heliconius* species and races (Jiggins *et al.* 2008), we also performed the comparison excluding these regions from the analysis. The Z chromosome was analyzed separately and compared with the autosomal genome. The mean of D_{xy} in F_{ST} peaks was significantly higher than in the background genome ($p=0.03$), however, removing the color pattern regions removed the difference as well ($p=0.29$). Mean π and number of fixed differences in F_{ST} if compared with the genomic background were significantly lower and higher respectively, regardless of the color pattern loci. The Z chromosome had significantly lower F_{ST} , D_{xy} , π and number of fixed difference relative to the autosomes (Fig. 6).

The estimation of F_{ST} accounts for both D_{xy} and π , we therefore illustrated the relationship between these two statistics, plotting the the distribution of D_{xy}/π across the genome. The plot of D_{xy} between *H. elevatus* and *H. pardalinus* corrected for π in *H. pardalinus* (Fig. 4) shows that D_{xy}/π perfectly concords with F_{ST} peaks. A similar plot of D_{xy}/π in *H. elevatus* gives almost the same result (Fig. S4).

Phylogeny of the regions of divergence

The alignment of all sites with $F_{ST} > 0.4$ concatenated produced the ML tree with mutual monophyly of both *H. elevatus* and *H. pardalinus* (Fig. 1C). This result was robust even when color pattern loci were excluded from the analysis. We also verified whether the F_{ST} peak tree of Fig. 1C

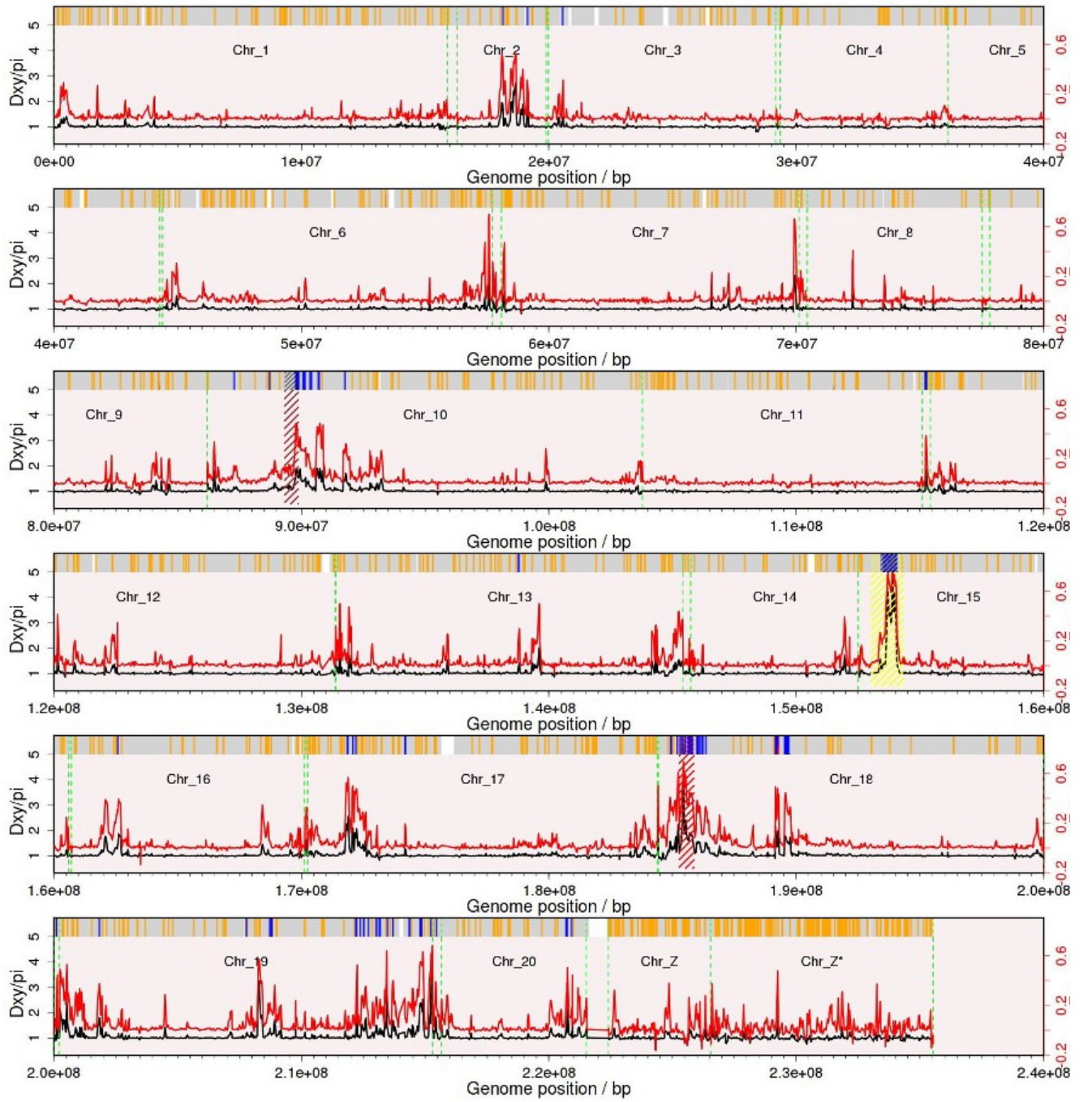


Figure 4. Genome scans from 50 kb sliding windows showing F_{ST} (red graph), D_{xy}/π (black graph), and phylogenetic information (track at top of the graph). Both D_{xy}/π and F_{ST} represent the divergence between Amazonian *H. elevatus* and *H. pardalinus*. D_{xy} was corrected for π in *H. pardalinus*. The phylogenetic information is coded by orange for the "genome tree," i.e. the genome-wide average tree of Fig. 2A; blue for the "species tree," i.e. complete mutual monophyly of both *H. pardalinus* and *H. elevatus* including all geographic populations of each; grey for all other trees; white stands for ingroup non-monophyletic, or missing data. Shaded regions represent the *Ac*, *N/Yb*, and *B/D* colour pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and named by «Chr...». The unmapped scaffolds of every chromosome are situated between the dash lines at the end of every chromosome. (The phylogenetic signal on this figure was visualized by N. Rosser and K. Dasmahapatra)

uniting the traditional geographic races of each species species could be the result of random fluctuations in coalescence between different regions by analyzing the phylogenies of the same F_{ST} peak regions of individual chromosomes separately. Suppose we have four taxa: (1) Amazonian

(Ecuador + Peru) *H. elevatus*, (2) allopatric *H. e. bari* from French Guiana, (3) Amazonian (Ecuador + Peru) *H. pardalinus*, and (4) allopatric *H. p. sergestus* from the Upper Huallaga/Mayo valley system in Peru. Then there are 15 possible rooted trees, only one of which shows mutual monophyly of the two species (((*H. p. sergestus*, Amazonian *H. pardalinus*), (*H. e. bari*, Amazonian *H. elevatus*)), outgroup) (hereafter the “species tree”). Chromosomes were used as independent data points; different peaks within chromosomes may be in linkage disequilibrium or in inversions which could cause some non-independent segregation of variation, but given that the chromosome numbers of *H. elevatus* and *H. pardalinus* are the same as for *H. melpomene* (Brown 1981), and in view of the overall conservation of synteny across the Lepidoptera (Heliconius Genome Consortium 2012), different chromosomes provide relatively independent markers of history. In our results, of 13 chromosomes having peaks with $F_{ST} > 0.4$, six provided evidence for mutual monophyly of *H. elevatus* (including *H. e. bari*) and *H. pardalinus* (including *H. p. sergestus*) rooted by *H. melpomene*. Assuming a “star tree” null hypothesis, where each of the four ingroup taxa of Fig. 1C are equally related to each other, if this result was sampled at random from the 15 possible genealogies, this “species tree” would have a very low probability (binomial test $p = 10^{-4}$). Even if only three trees showed the observed pattern, the relatively high frequency of such trees would remain unlikely ($p = 0.051$). Thus we are forced to a conclusion that the observed “species tree” topology in such a high fraction of F_{ST} peaks is unlikely to occur by chance.

In addition, we also checked how this “species tree” is distributed across the genome by classifying the previously obtained trees from the 50kb sliding window phylogenetic analysis into “species trees”, genome-wide trees and all other trees and plotting them across the genome (Fig. 4). This plot demonstrates that “species tree” mostly occurs in the regions of F_{ST} peaks, while the genome-wide trees are more or less evenly distributed in the regions of flat F_{ST} . It was particularly noticeable that the genome-wide tree is relatively more frequent in the Z chromosome.

Test for gene flow

D-statistics, a modified version of the four-population test (Reich *et al.* 2009) by Patterson *et al.* (2012), measuring the correlation of the allele frequencies between different lineages was used to test for the presence of gene flow between *H. pardalinus* and *H. elevatus*. The *D* was estimated with the assumed historical phylogeny ((*H. p. ssp. nov.*, *H. p. butleri*, *H. p. orteguaza*), *H. p. sergestus*), (*H. e. bari*, (*H. elevatus*, *H. e. pseudocupidineus*))). The plot of *D* values (Fig. S7) across the genome shows strong deviation from zero into positive side (average $D = 0.5$) indicating considerable admixture between amazonian *H. pardalinus* and *H. elevatus*. A few narrow dips of *D* statistics, signs of absence or reduced level of admixture, often correspond to the regions of F_{ST} peaks. The evidence of

correlation between D -statistics and F_{ST} can be derived from the highly significant difference in means between values of D statistics in F_{ST} peaks and the genomic background (Fig. 5, Table S2). D -statistics were noticeably higher in the Z chromosome than in the autosomes and this difference was highly significant (Fig. 6, Table S3). Thus, there is evidence that the level of gene flow in the Z chromosome is even higher than in the rest of the genome.

Functional characterization of the genomic regions of divergence

Among 16203 genes known for *Heliconius*, the F_{ST} peak regions contain 188 (1,2%). This amount of genes is not significantly different from the amount of DNA covered by these regions (1.4%). Thus, the regions of differentiation are not gene-rich. Gene ontology information was retrieved for 88 (47%) genes located in F_{ST} peaks and 6368 (39%) genes in the whole genome. These numbers are relatively proportional and indicate that the annotation of F_{ST} peaks was not strongly biased. The list of genes situated in the regions of F_{ST} peaks is provided in the Supplementary Information. Among these genes the most over-represented was the gene family *UDP-glycosyltransferase* which contained 13 out of 52 genes of this family found in *Heliconius*. *UDP-glycosyltransferase* is of major importance in detoxification and elimination of xenobiotic compounds. We also would like to point out odorant binding protein *HmOBP20* and olfactory receptor *HmOR30*, genes known to be involved in chemosensation. The *Bombyx mori* gene *BmOBP20*, a homolog of *Heliconius HmOBP20* (Heliconius Genome Consortium 2012), was shown to be differentially over-expressed in male silkmoth antennae (Dani *et al.* 2011). While *Heliconius HmOR30* together with *HmOR31* are homologous to receptor *BmOR30* (Heliconius Genome Consortium 2012), which is expressed predominantly in silkmoth female antennae (Wanner *et al.* 2007). Recent study on *Heliconius* gustatory receptors (Briscoe *et al.* 2013) provide evidence that olfactory receptor *HmOR30* has indeed female specific expression, but in the legs. Its expression was also observed in antennae but without sex-bias. The 50 kb window containing *OR31*, another duplicate of *BmOR30*, also has relatively high F_{ST} (0.37). Nevertheless, genes involved in chemosensation were not over-represented in the regions of divergence as there are 42 *HmOBP* and 64 *HmOR* genes known for *Heliconius*.

A GO enrichment test identified a number of enriched GO terms (Supp. Info. 3). Each peak as a separated subset of the data was compared against the whole genome. The majority of enriched GO terms were related to various metabolic processes, transferase activities and biosyntheses. The following terms were the most interesting: “sensory reception of smell”, “olfactory receptor activity”, and “*Wnt* signaling pathways”. Both “sensory reception of smell” and “olfactory receptor activity” are found to be enriched on chromosome 19, the same chromosome where we found genes *OBP20* and *OR30*. “*Wnt* signaling pathways” is enriched on chromosome 10. This term is of interest as the gene encoding

the signaling ligand *WntA* was shown to be associated with wing color formation across the *Heliconius* genus (Martin *et al.* 2012). The enrichment for “*Wnt* signaling pathways” was found in the first peak on chromosome 10. We verified the position of the *WntA* gene according to Martin *et al.* (2012) and it turned out to be exactly in this F_{ST} peak. However, this gene was not named in the *Heliconius* reference genome v1.1 (Heliconius Genome Consortium 2012). It is also worth mentioning that the *WntA* genomic region is in perfect linkage with the Anterior cell spot locus (*Ac*) that controls forewings color pattern shape in the *H. melpomene/cydno* phylogenetic group (Martin *et al.* 2012).

Selection

We compared the region of divergence versus the genomic background in Amazonian *H. elevatus* and *H. pardalinus* using *Tajima's D* statistics that enable a test for deviation from neutral evolution (*Tajima's D* = 0). *Tajima's D* of *H. elevatus* and *H. pardalinus* was negative for both divergent regions and genomic background (Fig. 5). Removing the color pattern sites from the divergent regions changed the sign of *Tajima's D* to positive. *Tajima's D* of the genomic background was always significantly more negative than in the regions of divergence ($p < 0.05$ for *H. elevatus* and $p < 0.001$ for *H. pardalinus*) and only *Tajima's D* of the genomic background was significantly different from zero.

The negative *Tajima's D* in the genomic background signifies that these regions might be under strong purifying selection and/or that there is population size expansion. In contrast, the weakly positive *Tajima's D* in the non-color pattern divergent sites suggests balancing selection, which seems to be different from the near neutrality or possible purifying selection at the color pattern divergent sites. The latter however is vague as all the divergent sites are not really significantly different from *Tajima's D* being zero.

Tajima's D in the autosomal genome was consistent between species, while the Z chromosome showed some discrepancy. *Tajima's D* of the Z chromosome was significantly positive in *H. elevatus* and negative in *H. pardalinus*, but always significantly different from the autosomal *Tajima's D* (Fig. 6). Unlike in autosomes, *Tajima's D* in the Z chromosome was also significantly different between *H. elevatus* and *H. pardalinus* ($p < 0.001$). This result implies that *H. elevatus* is probably experiencing different selective pressure on its Z chromosome and/or it may be somehow related with gene flow. The positive *Tajima's D* indicates that the Z chromosome of *H. elevatus* is under balancing selection and/or its population size decreases, while the negative *Tajima's D* in *H. pardalinus* signifies that its Z chromosome along with autosomes might be under purifying selection and/or its population size expands.

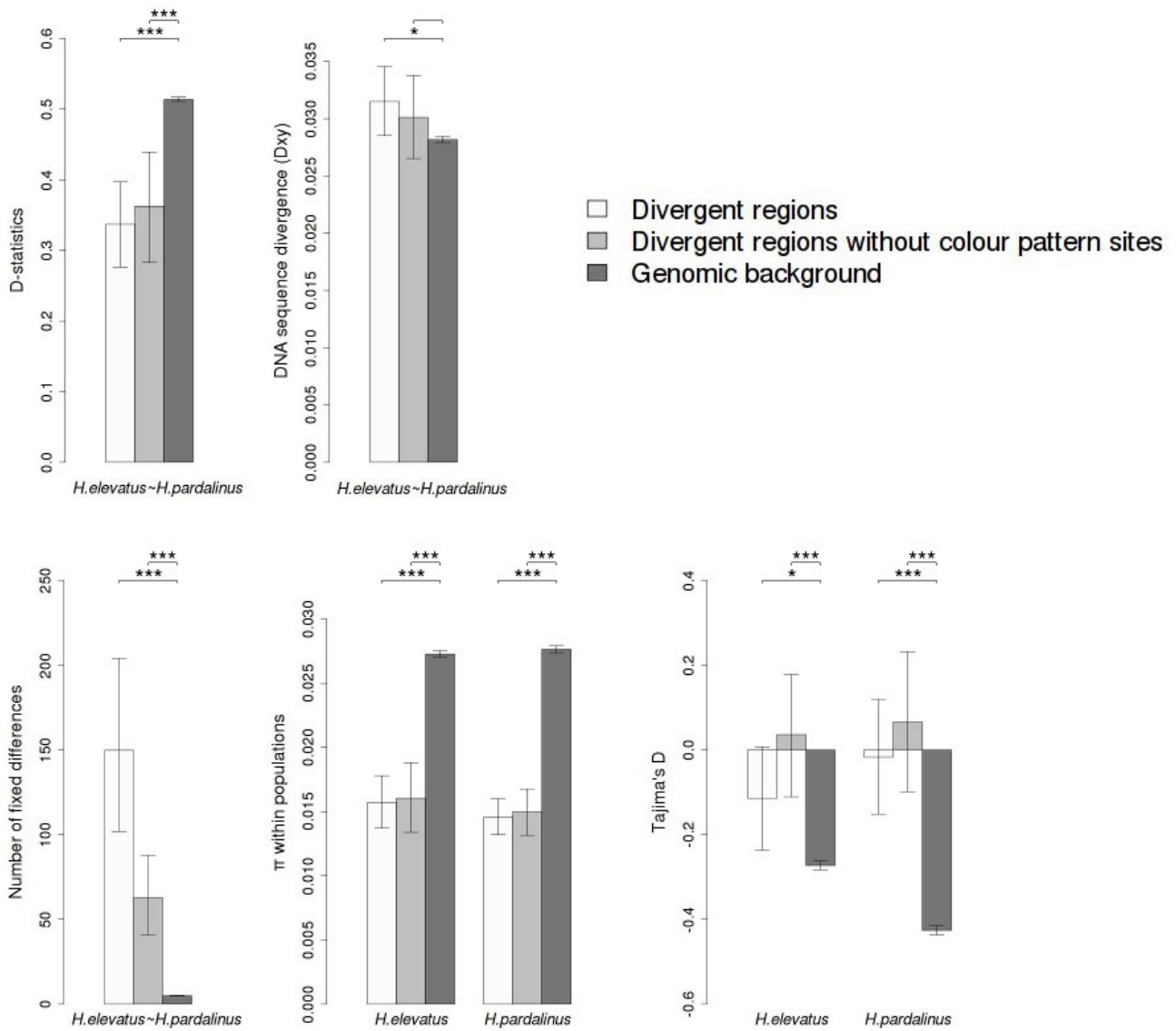


Figure 5. Comparison of mean D_{xy} , π , D -statistics, Tajima's D and number of fixed differences for the regions of species differentiation ($F_{ST} > 0.4$) vs. the rest of the genome. The 97.5 and 2.5 percentile confidence intervals and significance of the difference in means were computed by bootstrap resampling. The Z chromosome was excluded from the genomic background in this comparison. *, **, *** - indicate the level of significance of 0.05, 0.01, 0.001 respectively.

Investigations of several genes for the variation in the ratio of synonymous and non-synonymous substitutions ($\omega = dN/dS$) across the phylogeny identified that genes from the color-pattern regions - *optix*, *WntA*, and *penguin-like* - are highly conserved and under strong negative selection on the protein against amino-acid variants in all lineages ($\omega \ll 1$) (Table. S7). Only the *fizzy-like* gene *fzl* in *H. elevatus* had the ω significantly different from the other lineages ($\omega = 0.51$ vs. $\omega = 0.22$). This twice higher ω implies reduction of conservative selection in *fizzy-like*. The analysis of this gene in the branch-site model that allows for variation of the selective pressure across sites, reveals that there are five amino acids that are under strong positive selection ($\omega = 7.55$) (see Table. S8).

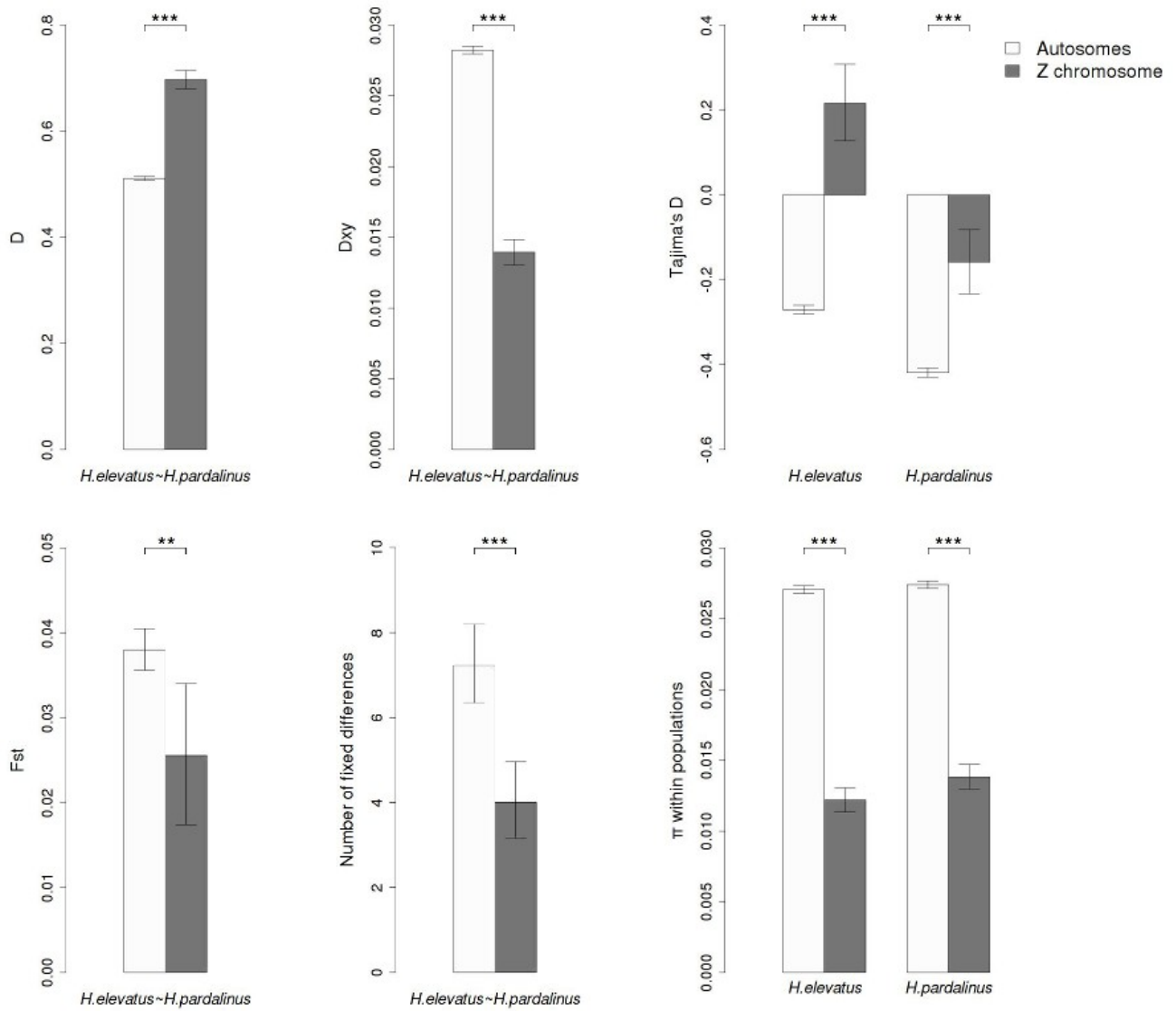


Figure 6. Comparison of mean F_{ST} , D_{xy} , π , Tajima's D and number of fixed differences for the Z chromosome vs. autosomes. The 97.5 and 2.5 percentile confidence intervals and significance of the difference in means were computed by 10000 bootstrap resampling. *, **, *** - indicate the level of significance of 0.05, 0.01, 0.001 respectively.

In addition to genes coding wing-patterning, we also inspected dN/dS ratio in several genes involved in chemosensation. *OBP20* and *OR30* were representatives of the regions of divergence, while *OBP29*, *OBP31* and *OR36*, genes from the same gene families, were representatives of the regions with very low differentiation (F_{ST} close to 0). Only the gene *OBP20* had the ω significantly greater than 1 in both *H. elevatus* and *H. pardalinus* ($\omega = 2.22$ and $\omega = 1.56$ respectively) implying excess of non-synonymous variation. While analysis of *OBP29*, *OBP31*, *OR36* and *OR30* found no difference between *H. elevatus*, *H. pardalinus* with $\omega < 1$ (Table. S7).

Discussion

Data quality

In spite of the fact that this study is focused on representatives of the silvaniform group that is quite divergent from the cydno/melpomene (Heliconius Genome Consortium 2012), and the silvaniform reference genome is not available yet, we obtained data which is comparable or even exceed quantitatively and qualitatively those used in a recent whole genome analysis of *Heliconius* (Martin *et al.* 2013). For example, with very similar filtering and alignment parameters, our nuclear genome alignment, which comprises 32 genomes of the silvaniforms and 3 genomes from the cydno/melpomene group, covers 90% of the reference genome, while in the paper by Martin *et al.* (2013), where among 31 genomes the majority were from the cydno/melpomene group, the average coverage of the reference genome was 70%. Unfortunately, our data was more biased by indels compared with studies on the cydno/melpomene group. Because of the divergence of the silvaniforms from *H. melpomene* reference genome, indels in either reference genome or in the species we analyze here are not distinguishable from missing data due to sequencing failure. Nevertheless, we retrieved more complete sites (66 Mb) than Martin *et al.* (2013) did (60 Mb). A better alignment was obtained for the mitochondrial genome as well. Keeping only complete sites in the alignment over all species, we covered 78% of the *H. melpomene* mitochondrial reference genome, while Martin *et al.* (2013) obtained the alignment of only 65% of the reference genome (9.5 kb).

When interspecific admixture masks species phylogeny

We complemented the previously reported whole genome phylogeny of *H. pardalinus* and *H. elevatus* (Heliconius Genome Consortium 2012) by including additional races of these species. The topology of the new phylogenetic tree (Fig. 2A) indicates that both *H. elevatus* and *H. pardalinus* are rather paraphyletic in terms of the genome-wide phylogeny. According to the phylogenetic species concept, it might be concluded that there are actually four species rather than two. The mitochondrial DNA also shows evidence of admixture or polymorphism among the *Heliconius* species (Fig. 2B). Previous studies have shown strong divergence at mitochondrial loci even between closely related pairs of *Heliconius* species (Brower 1996; Beltrán *et al.* 2007), as expected given the tendency of sterility of female hybrids (Haldane's Rule) in *Heliconius* and the Lepidoptera in general (Sperling 1990). However, these two species have been well recognized by their differences in color pattern and genitalic morphology for almost half a century (Emsley 1965; Brown 1976; Lamas *et al.* 2004). In addition, recent study of RAD sequences of several *Heliconius* species (Heliconius Genome Consortium 2012) has revealed that *H. elevatus* has introgressions in the *B/D* and *N/Yb/P* color-pattern

loci from *H. melpomene*. This introgression has direct influence on phenotype and is assumed to play important role in isolation of *H. elevatus* from *H. pardalinus*. Hence, at least all races of *H. elevatus* should be monophyletic, but this is not what we observe.

However, the original morphology-based taxonomy might still be trustworthy under the apparently unlikely scenario that the whole genome phylogeny has been totally distorted by extensive admixture. A more detailed investigation reveals that the high level of admixture between *H. elevatus* and *H. pardalinus* is actually very probable. First of all, it is known that sympatric members of the entire melpomene-silvaniform group of *Heliconius*, including these two species, undergo occasional hybridization and backcrossing across all ~15 species (Mallet *et al.* 2007). Secondly, the topology of the obtained whole genome phylogenetic tree is consistent with gene flow between sympatric Amazonian species. Ecuadorian and Peruvian *H. elevatus* and *H. pardalinus* are clustered together, *H. p. sergestus* residing not far from them but staying geographically isolated is at the base of this group, and *H. e. bari* from French Guiana, which is on the opposite side of the continent is an outgroup for all of them together. Thus, hybridization in the Amazon led to convergence of local races of *H. elevatus* and *H. pardalinus*, while geographic isolation of *H. p. sergestus* and *H. e. bari* allowed them to diverge from the Amazonian group. Our analysis of the phylogenetic signal across the genome using the sliding window approach supports this hypothesis (Fig 4). There are actually not that many trees across the genome which support the obtained whole genome tree (see branch support Fig. 2A, and there are even less trees which support the morphology-based splitting history (Fig. 4). Furthermore, blurred phylogenetic relationships within the Amazonian group (Fig. 3) illustrate that admixture between *H. elevatus* and *H. pardalinus* is not only possible, but that these species might actually be close to panmixia.

How much of the genome is admixed?

A whole genome analysis of *H. timareta* and *H. cydno* has revealed that these species exchange up to 40% of their genomes with sympatric *H. melpomene* (Martin *et al.* 2013). However, in spite of this high level of admixture, the whole genome phylogenetic signal was still consistent with the accepted species splitting history. Following the same logic as in Martin *et al.* (2013), from the frequency of different trees across the genome, we can conclude that the level of admixture between *H. elevatus* and *H. pardalinus* must be far higher. The obtained pattern of F_{ST} (Fig. 3) implies that the fraction of the genome which remains differentiated between the two species is around several percents. The most differentiated regions coincide with the color pattern loci *B/D* and *N/Yb/P*, which are known to show a phylogenetic signal suggesting transference of the “dennis-ray” pattern from *H. melpomene* to *H. elevatus* (Heliconius Genome Consortium, 2012), and so explaining the major distinctive color

pattern differences between *H. elevatus* and *H. pardalinus* (Fig. 2, 3). Thus at least this evidence from color pattern accords with the idea that loci determining the traits differing between species are within these F_{ST} peaks, or "islands of genomic divergence." It is also noteworthy that the obtained background F_{ST} level ($F_{ST} < 0.4$) is unusually low (mean = 0.03) and is comparable to the F_{ST} between the freely hybridizing sympatric races of *H. melpomene* (Heliconius Genome Consortium 2012; Martin *et al.* 2013). Hence, it can be concluded that there is indeed rampant gene flow between *H. elevatus* and *H. pardalinus* with very few regions maintaining species differences.

The statistic F_{ST} measures divergence (D_{xy}) relative to within-species levels of polymorphism (π). A plot of absolute divergence D_{xy} on its own shows little evidence of the relative divergence peaks found with F_{ST} (Fig. S5). Such evidence has been regarded as *de facto* evidence that genomic islands of divergence as detected by F_{ST} could be artifacts of nucleotide diversity reduction trough within one or the other of a pair of completely reproductively isolated species (Turner & Hahn 2010). Close investigation of the pattern of divergence revealed that the mean D_{xy} of F_{ST} peaks is significantly higher than in the background genome inferring that though regions of F_{ST} peaks are not the highest in the D_{xy} plot, they are placed among high D_{xy} peaks. However, this is mainly due to divergence in the color pattern regions as removing these regions from the analysis leads to insignificant difference between the sites of divergence and genomic background (Fig. 5). On the other hand, comparison of the π of the F_{ST} peaks and the background genome shows that the π is significantly lower in the regions of divergence regardless of the inclusion of the color pattern loci into the analysis (Fig. 5). Hence, correspondence between D_{xy} and F_{ST} is not noticeable due to the considerably lower polymorphism level in the regions of divergence. A plot of D_{xy} between *H. elevatus* and *H. pardalinus* corrected for π is consistent with this conclusion (Fig. 4, S4). The regions of F_{ST} peaks are also significantly richer in numbers of fixed differences (Fig. S6). Of course, such statistics as F_{ST} , D_{xy} , π and fixed differences are not independent, but they all are concordant in our work, provide support for the model of "genomic islands of speciation" and illustrate that these "islands of speciation" can be unimaginably small.

Evidence for gene flow

A high level of homogenization across the genome with several regions of divergence can also be explained by shared ancestral polymorphism when *H. elevatus* and *H. pardalinus* are in fact completely reproductively isolated. Ancestral polymorphism is a major argument against hybrid speciation in *Heliconius* by Brower (2011, 2012) and it is discussed in studies of *Anopheles* (Turner & Hahn 2010; Hahn *et al.* 2012). One approach to reveal gene flow between species is to perform a geographically controlled test (Grant *et al.* 2005; Martin *et al.* 2013). An excess of shared alleles in

sympatry would indicate recent gene flow. Tests of the phylogenetic structure were also successfully used to reveal recent gene flow (Reich *et al.* 2009, 2012; Green *et al.* 2010; Patterson *et al.* 2012; Martin *et al.* 2013). We performed a geographically controlled test of phylogenetic structure estimating D -statistics (Patterson *et al.* 2012) inferred from the four-population test (Reich *et al.* 2009, 2012) and found that D -statistics do suggest a major influence of admixture on the genome (Fig. 5). Importantly, the regions of low D implying reduced gene flow are usually associated with regions of F_{ST} peaks (Fig. 6, S7). The estimation of D -statistics relies on solid knowledge of history based on a genome-wide tree as a surrogate for the major historical branching pattern. However, in the current study the genome-wide average tree is so strongly distorted by admixture that we can no longer infer the original branching pattern from its topology. We therefore used the traditional morphology-based phylogenetic tree for this test, *a priori* assuming that it represents the true species history.

Another way to test for gene flow was to check if the F_{ST} peak regions (Fig. 4) represent the species-determining regions of the genome. As we expected, the concatenated F_{ST} peak regions produced a “species tree”, where *H. elevatus bari* clusters together with Amazonian races of *H. elevatus* and *H. p. sergestus* joins the other *H. pardalinus* in spite of their allopatry. Analysis of F_{ST} peaks by chromosome revealed that 6 chromosomes out of 13 having F_{ST} peaks showed the “species tree”. This observation is unlikely by chance ($p=10^{-4}$) and confirms that this tree is not the result of a random fluctuation in coalescence between different regions. Thus, our decision to use the morphology-based phylogenetic tree in estimating D -statistics was in some sense correct.

The observed pattern of nucleotide polymorphism across the genome was also consistent with the presence of the gene flow. The π within the F_{ST} peaks is close to the half that of the π in the genomic background (Fig. 5). This is almost exactly as expected if both species share variation at the background regions, so that the effective population size is approximately doubled. The neutral expectation of π will be equal to $4Ne\mu$, where Ne is effective population size, and μ is the per site mutation rate. So if Ne is doubled, the π should get doubled as well.

The comparison of the variation in low and high frequency polymorphisms in the region of divergence and genomic background using *Tajima's D* is also consistent with the hypothesis of rampant gene flow between *H. elevatus* and *H. pardalinus*. The obtained significantly negative *Tajima's D* in the genomic background can be considered to be an indication of population size expansion. If the level of gene flow between the two species is as high as to be approaching panmixia, it virtually increases the population size. Besides, given the near identity of *Tajima's D* between the Amazonian species at these regions (Fig. 5), it signifies that background selective sweeps that remove rare deleterious variants are working between as well as within species, and may also explain a lot of the genome homogenization.

The last, but not the least, evidence of gene flow is that *H. elevatus* and *H. pardalinus* do hybridize and mature. Recent crosses in captive populations produced viable progeny (Dasmahapatra & Rosser, in prep.). However, the F2 generation has yet to be obtained and any conclusions about hybrid fertility cannot be made at the moment.

What maintains species boundaries?

H. pardalinus and *H. elevatus* are widely sympatric and if their genetic differences are limited to only a few percent of the genome, these regions are probably the genomic regions that maintain species differences. The most obvious character separating the two species is the radical difference in color pattern (Fig. 2) and mimicry, with *H. elevatus* mimicking other rayed *Heliconius*, such as *H. erato* and *H. melpomene*, and *H. pardalinus* mimicking *Melinaea* and other "tiger-pattern" ithomiines and heliconiines. This phenotypic difference can be explained by divergence in the main color pattern loci *Ac*, *B/D* and *N/Yb/P* on chromosomes 10, 15 and 18 respectively. The latter two loci are from the two highest F_{ST} peaks (Fig. 4). While locus *Ac* is not precisely within the F_{ST} peak, it is very close to the peak of the tightly linked *WntA* locus (Martin *et al.* 2012) that is within the F_{ST} peak. The F_{ST} signal in these loci is further supported by the fact that *H. elevatus* has genetic introgressions from *H. melpomene* in *B/D* and *N/Yb/P*, and all *H. pardalinus* races possess an inversion within *N/Yb/P* (Joron *et al.* unpublished) identical to one of those found in *H. numata* (Joron *et al.* 2011). As color pattern plays a key role in Müllerian mimicry and mate preference (Jiggins *et al.* 2004; Mavarez *et al.* 2006; Kronforst *et al.* 2006; Merrill *et al.* 2011b; a), these three "supergenes" are indeed the probable factors maintaining species differences.

In addition, there are other regions of high F_{ST} and they may be as important as wing-patterning loci. We discovered that these regions contain some chemosensation genes (*OBP20*, *OR30*) from odorant binding protein and olfactory receptor families. Chemosensation is probably the second most important trait in *Heliconius* that is important for interspecific ecological differences. Olfactory receptor *HmOR30* was shown to be female-specifically expressed in legs (Briscoe *et al.* 2013) and can be assumed to be involved in host-plant recognition. There are no studies on the expression of *Heliconius* odorant binding proteins, but *OBP20* is most probably male-specific as this gene is homologous to the *B. mori* gene *BmOBP20* that is expressed only in male antennae (Dani *et al.* 2011). Hence, *OBP20* can be related to male recognition of a conspecific female during courtship. The importance of these genes in ecological divergence between *H. pardalinus* and *H. elevatus* is also supported by the enrichment for the GO terms "sensory reception of smell" and "olfactory receptor activity" in the regions of divergence. Nevertheless, it is very difficult at this stage to make unambiguous conclusions about ecological differences between *H. pardalinus* and *H. elevatus* based

on these genes only, as these gene families comprise many genes (Heliconius Genome Consortium 2012) and very little is known about their relative importance. Further bioassay work is required to test the involvement of these candidate genes in speciation.

Species differences in the color-pattern region and chemosensation were also supported by the dN/dS ratio analysis. The sites under positive selection in the *OBP20* gene are different in *H. elevatus* and *H. pardalinus*. *H. elevatus* has sites under positive selection in *fizzy-like* (Table S8), a gene considered to be important in the *N/Yb/P* color pattern region (Jiggins et al. Unpublished), while *H. pardalinus* does not show a significantly different dN/dS ratio at this gene compared with other taxa investigated in this study. The absence of divergence in dN/dS ratio of other genes implies that F_{ST} peaks are also due to some non-coding differences. However, this does not mean that these genes are not important in species divergence. For example, our study is in concordance with the previous work (Reed et al. 2011) and provides evidence that amino acid composition of *optix* is highly conserved. However, the expression of this gene was shown to be controlling red color wing-patterning in *Heliconius* and its functional variation was suggested to be due to *cis*-regulation in an adjacent non-coding region (Reed et al. 2011). Thus, there is still much to discover in gene action and we believe that more extensive analyses for selection and future functional characterization of *Heliconius* genes will help to understand the identified genomic regions of divergence better.

Admixture in the Z chromosome

In *Heliconius* butterflies, as in the rest of Lepidoptera and in birds, the female sex is heterogametic. The Z chromosome is considered to diverge faster than autosomes in the neutral scenario given its somewhat reduced effective population size (Whitlock & Wade 1995). There are also theoretical expectations and preliminary empirical evidence that sexual traits and genes of prezygotic isolation should be disproportionately Z-linked and that they should enhance the divergence of the Z chromosome (Qvarnström & Bailey 2008). In accord with these predictions and studies showing stronger sex chromosomal differentiation among closely related species (Garrigan et al. 2012; Ellegren et al. 2012; Lee et al. 2013) it was shown that the Z chromosome in *Heliconius* was more divergent between *H. melpomene* and *H. cydno* than the autosomes (Kronforst et al. 2013; Martin et al. 2013). Furthermore, the Z chromosome harbors genes linked to hybrid female sterility that contribute to Haldane's Rule (Naisbit et al. 2002).

In strong contrast to prior evidence, our own results suggest that the whole Z chromosome is less divergent than the autosomes between *H. pardalinus* and *H. elevatus* in the Amazon. The Z chromosome shows considerably lower levels of genomic divergence (F_{ST} , D_{xy} , fixed differences) compared to autosomes. There are some small peaks of F_{ST} on the Z chromosome, these are however

narrow and may be artifacts of lower levels of polymorphism than on the autosomes. The predominance of the genome tree topology in Z chromosomal windows (Fig. 4), and the stronger *D*-statistics also provide evidence that the Z chromosome is subject to stronger admixture than the rest of the genome (Fig. 5). This phenomenon together with the phylogenetic pattern of the mitochondrial genome would be difficult to explain if Haldane's Rule of hybrid female sterility holds between *H. pardalinus* and *H. elevatus*.

Conclusion

Recent theoretical and empirical work on ecological speciation has questioned many established paradigms. Such phenomena as speciation with gene flow (e.g. Drès & Mallet 2002; Emelianov et al. 2004; Nosil 2008), hybrid speciation (reviewed in Mallet 2007), adaptive introgression (reviewed in Mallet 2005) and even up to 40% interspecific genomic admixture (Martin *et al.* 2013) are no longer considered unlikely. The work described here further complement these recent discoveries by suggesting that species can remain distinct in spite of occasional hybridization and backcrossing leading to almost total admixture between species. This admixture is so strong that it distorts the phylogenetic topology of the majority of the genome away from the original species history. It is important to note that *H. pardalinus* and *H. elevatus* are "good species," which show no evidence of the genomic admixture in their appearance or behavior: the species are not cryptic, and have never before been suspected even to be particularly close relatives. The current authors investigated this pair because of paraphyly found during preliminary studies associated with the *H. melpomene* genome project (Heliconius Genome Consortium, 2012). Given our finding that these two species can be maintained in the face of previously unimaginable levels of admixture, it seems likely that speciation with gene flow could be considerably more common than previously realized.

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Author contribution

This work was carried out mostly by me. Reads alignment to the reference genome, removing of PCR duplicates and indels realignment were performed without my direct involvement by Kanchon Dasmahapatra. Neil Rosser and Kanchon Dasmahapatra visualized the phylogenetic signal across the genome and I presented their result in Fig. 4. The outline of the thesis was produced with a great contribution of James Mallet.

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Supplementary Information

Table S1. Samples information

Species	Sex	Location	Coordinates	Sequencing coverage x
<i>H. elevatus pseudocupidineus</i>	male	Idea Religiosa, Munichis Loreto Peru	05°54'36.7" S 76°13'33.4" W	29
<i>H. elevatus pseudocupidineus</i>	male	Km-75 Tarapoto-Yurimaguas San Martín Peru	06°10'37.0" S 76°15'24.4" W	37
<i>H. elevatus pseudocupidineus</i>	female	Km-103.1 Tarapoto-Yurimaguas-road to Micaela Bastida, 2.5 km in Loreto Peru	05°58'18.2" S 76°13'55.2" W	36
<i>H. elevatus pseudocupidineus</i>	female	Biodiversidad UNSM, Km-17.2 Tarapoto-Yurimaguas San Martín Peru	06°27'42.6" S 76°17'30.8" W	25
<i>H. pardalinus ssp. nov.</i>	male	Caño Tushmo, Lago Yarinacocha, Pucallpa Ucayali Peru	08°20'32.7" S 74°35'31.8" W	26
<i>H. pardalinus ssp. nov.</i>	female	Caño Tushmo, Lago Yarinacocha, Pucallpa Ucayali Peru	08°20'32.7" S 74°35'31.8" W	33
<i>H. pardalinus ssp. nov.</i>	female	Caño Tushmo, Lago Yarinacocha, Pucallpa Ucayali Peru	08°20'32.7" S 74°35'31.8" W	31
<i>H. pardalinus ssp. nov.</i>	male	Caño Tushmo, Lago Yarinacocha, Pucallpa Ucayali Peru	08°20'32.7" S 74°35'31.8" W	37
<i>H. elevatus bari</i>	male	Route de Regina, Morne aux Echos Guyane Francaise	x	32
<i>H. elevatus bari</i>	female	Patawa Laie#3 Guyane Francaise	x	36
<i>H. elevatus bari</i>	male	Patawa Laie#3 Guyane Francaise	x	28
<i>H. elevatus bari</i>	female	Patawa Laie#3 Guyane Francaise	x	29
<i>H. pardalinus butleri</i>	female	Pongo-Barranquita km3.5 (Neil 544) San Martín Peru	x	27
<i>H. pardalinus butleri</i>	female	Tarapoto-Yurimaguas km 59.3 San Martín Peru	x	23
<i>H. pardalinus butleri</i>	male	Idea Religiosa, Munichis Loreto Peru	05°54'36.7" S 76°13'33.4" W	34
<i>H. pardalinus butleri</i>	male	Idea Religiosa, Munichis Loreto Peru	05°54'36.7" S 76°13'33.4" W	32
<i>H. pardalinus sergestus</i>	female	Tarapoto-Urahuasha trail San Martín Peru	06°28'39.8" S 76°21'05.9" W	33
<i>H. pardalinus sergestus</i>	male	Tarapoto-Urahuasha trail San Martín Peru	06°28'39.8" S 76°21'05.9" W	32
<i>H. pardalinus sergestus</i>	female	Tarapoto-Urahuasha trail San Martín Peru	06°28'39.8" S 76°21'05.9" W	29
<i>H. pardalinus sergestus</i>	male	Tarapoto-Urahuasha trail San Martín Peru	06°28'39.8" S 76°21'05.9" W	35
<i>H. pardalinus orteguaza</i>	male	Ecuador: Orellana, Anangu	x	38
<i>H. pardalinus orteguaza</i>	male	Ecuador: Orellana, Anangu	x	24
<i>H. pardalinus orteguaza</i>	male	Ecuador: Orellana, Anangu	x	34
<i>H. pardalinus orteguaza</i>	female	Ecuador: Orellana, Anangu	x	20
<i>H. elevatus</i>	male	Apuya Km2.7 Ecuador	S1 06.934 W77 46.700	30
<i>H. elevatus</i>	male	El Capricho Km 5.7 Ecuador	S1 11.269 W77 49.86	19

<i>H. elevatus</i>	male	Km31 Com Cajabamba Ecuador	S1 20.290 W77 50.123 954	32
<i>H. elevatus</i>	female	Km35 Sendero PtaSanCristobal Ecuador	S1 17.446 W77 50.514	22
<i>H. hecale felix</i>		Peru		34
<i>H. hecale felix</i>		Peru		26
<i>H. hecale felix</i>		Peru		38
<i>H. hecale felix</i>		Peru		29
<i>H. melpomene aglaope</i>				42
<i>H. melpomene amaryllis</i>				43
<i>H. timareta ssp. nov.</i>				44

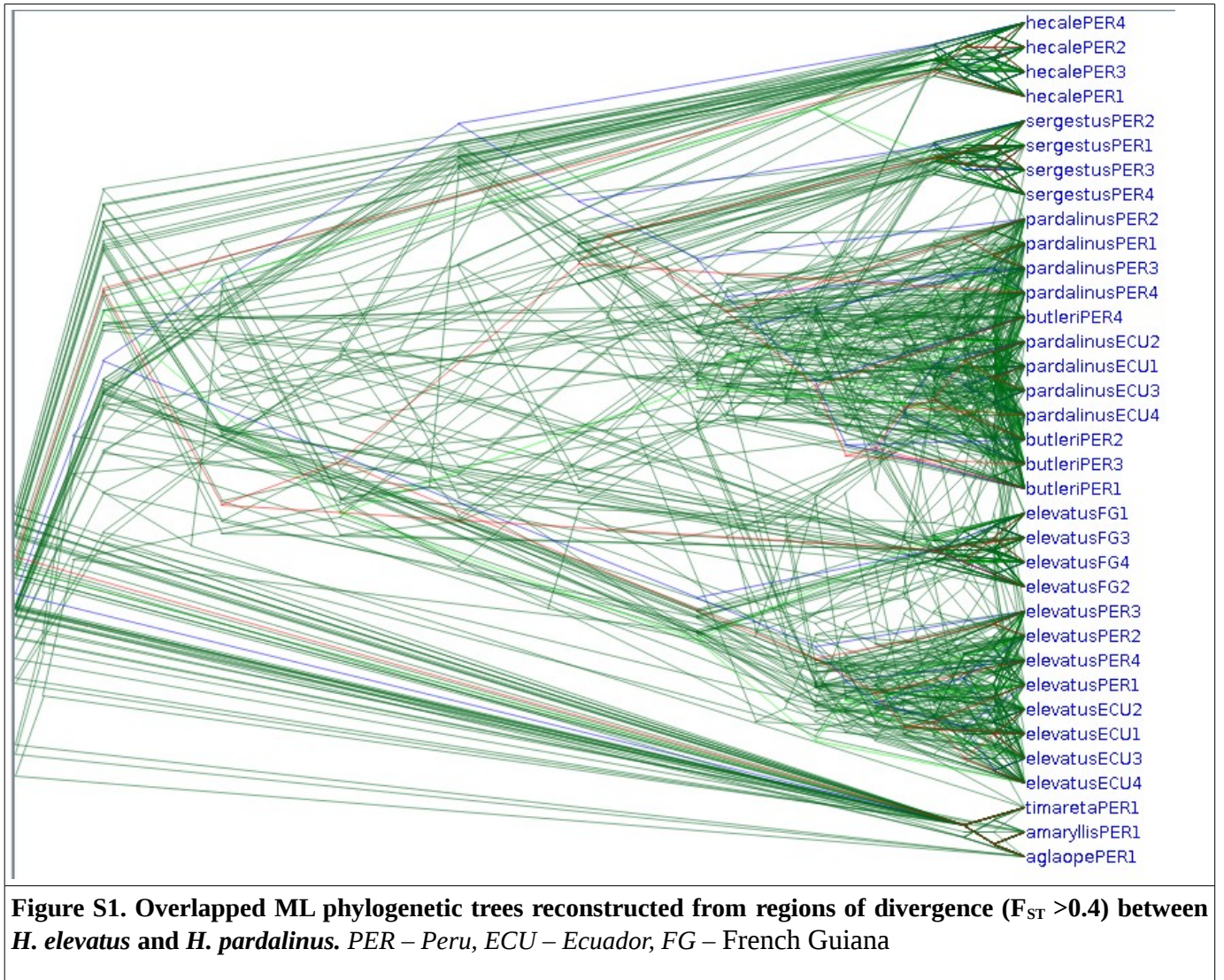
Table S2. Comparison of mean D_{xy} , π , D -statistics, $Tajima's D$ and number of fixed differences for the regions of divergence ($F_{ST} > 0.4$) vs. the genomic background.

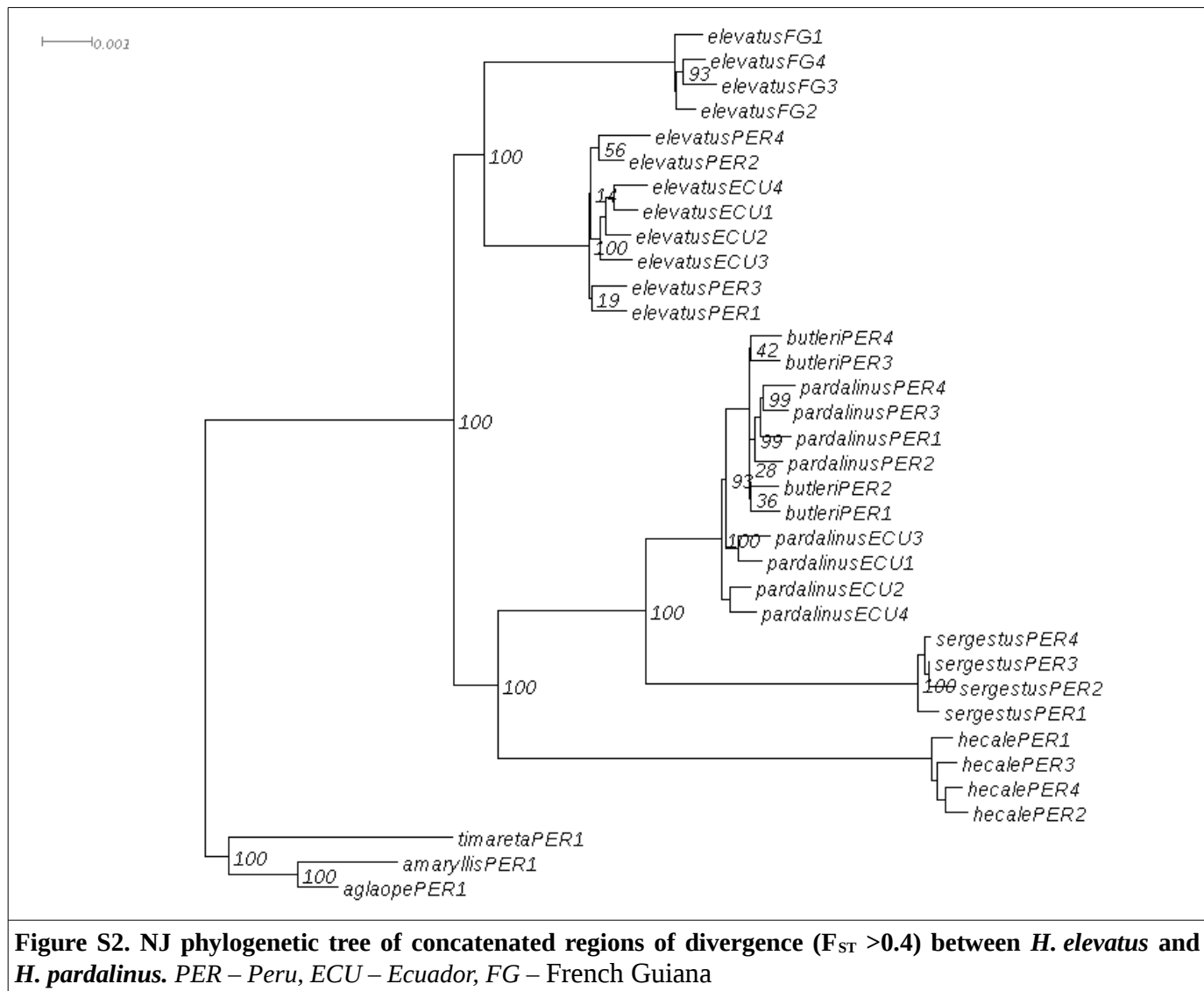
Mean		Confidence intervals		p-value
Regions of diverg.	Background	Regions of divergence	Background	
Dxy				
with color pattern. regions				
0.03151194	0.02818285	0.02853716 0.03457164	0.02792027 0.02844373	0.0295
without color pattern. regions				
0.03011702	0.02818492	0.02653191 0.03378516	0.02792564 0.02844844	0.2935
D-statistics				
with color pattern. regions				
0.3368798	0.5138098	0.2762835 0.3973252	0.5104553 0.5171427	0.000
without color pattern. regions				
0.3626597	0.5142375	0.2828240 0.4393366	0.5109328 0.5176486	1e-04
Fixed differences				
with color pattern. regions				
150	5	101.8642 204.1649	4.868056 5.250233	0.000
without color pattern. regions				
62	5	40.89255 87.83032	4.809120 5.186984	0.000
π H. elevatus				
with color pattern. region				
0.0157	0.02725299	0.01373429 0.01779403	0.02698466 0.02751971	0.000
without color pattern. regions				
0.01604894	0.02727119	0.01337660 0.01879585	0.02700716 0.02753605	0.000
π H. pardalinus				
with color pattern. region				
0.01455672	0.02764426	0.01318955 0.01600004	0.02738085 0.02790930	0.000
without color pattern. regions				
0.01498085	0.02764426	0.01315947 0.01676170	0.02737481 0.02790499	0.000

Tajima's D <i>H. elevatus</i>				
<i>with color pattern. region</i>				
-0.113909	-0.2728717	-0.237012649 0.007054739	-0.2830720 -0.2626224	0.0111
<i>without color pattern. regions</i>				
0.03625532	-0.2727528	-0.1105303 0.1779221	-0.2832148 -0.2621773	0.000
Tajima's D <i>H. pardalinus</i>				
<i>without color pattern. regions</i>				
-0.01684776	-0.4258957	-0.1513312 0.1200709	-0.4363358 -0.4151308	0.000
<i>without color pattern. regions</i>				
0.06605957	-0.4258957	-0.09837926 0.23191074	-0.4364624 -0.4152341	0.000

Table S3. Comparison of mean F_{ST} , D_{xy} , π , Tajima's D and number of fixed differences for the Z chromosome vs. autosomes.

Mean		Confidence intervals		p-value
Autosomes	Z chromosome	Autosomes	Z chromosome	
<i>F_{ST}</i>				
0.03803401	0.02558127	0.03553330 0.04054469	0.01755713 0.03393185	0.0039
<i>D_{xy}</i>				
0.02823283	0.01394794	0.02797447 0.02849418	0.01304454 0.01484083	0.000
D-statistics				
0.5111537	0.6975456	0.5077603 0.5146679	0.6800464 0.7146202	0.000
fixed differences				
7.224065	4	6.349294 8.196796	3.164794 4.958801	0.000
<i>π H. elevatus</i>				
0.02707885	0.01219775	0.02681051 0.02734728	0.01133707 0.01306368	0.000
<i>π H. pardalinus</i>				
0.02742039	0.01382434	0.01294418 0.01469404	0.02715520 0.02767799	0.000
<i>Tajima's D H. elevatus</i>				
-0.2704756	0.21603	-0.2806184 -0.2601577	0.1272119 0.3077979	0.000
<i>Tajima's D H. pardalinus</i>				
-0.418812	-0.1584273	-0.4294652 -0.4079417	-0.23285561 -0.08200698	0.000





0.001



Figure S3. NJ phylogenetic tree of concatenated regions of divergence ($F_{ST} > 0.4$) between *H. elevatus* and *H. pardalinus* with color pattern regions excluded PER – Peru, ECU – Ecuador, FG – French Guiana

Table S7. Test for difference in evolutionary rate using a branch model in PAML

Model	ω_0	ω_p	ω_e	lnL	$2\Delta L (\chi^2)$
<i>OBP20</i> (within the regions of divergence chr 19)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.54	$=\omega_0$	$=\omega_0$	-819.43	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.40	2.22	$=\omega_0$	-817.76	3.33
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.45	$=\omega_0$	1.56	-818.68	1.50
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.29	2.22	1.56	-816.36	6.13*
<i>OR30</i> (within the regions of divergence chr 19)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.10	$=\omega_0$	$=\omega_0$	-2205.28	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.08	0.15	$=\omega_0$	-2204.45	1.66
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.11	$=\omega_0$	0.09	-2205.25	0.06
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.08	0.14	0.09	-2204.42	1.72
<i>penguin-like</i> (within the regions of divergence chr 15)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.13	$=\omega_0$	$=\omega_0$	-3170.47	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.12	0.15	$=\omega_0$	-3170.37	0.2
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.15	$=\omega_0$	0.06	-3169.27	2.4
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.15	0.15	0.06	-3169.27	2.4
<i>WntA</i> (within the regions of divergence chr 10)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.0001	$=\omega_0$	$=\omega_0$	-1569.09	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.0001	0.0001	$=\omega_0$	-1569.09	0
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.0001	$=\omega_0$	0.0001	-1569.09	0
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.0001	0.0001	0.0001	-1569.09	0
<i>fzl</i> (within the regions of divergence chr 15)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.27	$=\omega_0$	$=\omega_0$	-2872.63	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.31	0.20	$=\omega_0$	-2871.54	2.18
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.22	$=\omega_0$	0.51	-2870.08	5.1*
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.24	0.20	0.51	-2869.86	5.54
<i>optix</i> (within the regions of divergence chr 18)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.07	$=\omega_0$	$=\omega_0$	-1481.13	
<i>H1</i> : $\omega_p \neq \omega_0 = \omega_e$	0.08	0.06	$=\omega_0$	-1480.91	0.44
<i>H2</i> : $\omega_p = \omega_0 \neq \omega_e$	0.06	$=\omega_0$	0.11	-1480.46	1.34
<i>H3</i> : $\omega_p \neq \omega_0 \neq \omega_e$	0.06	0.06	0.11	-1480.44	1.38
<i>OBP29</i> (within the genomic background chr 9)					
<i>H0</i> : $\omega_p = \omega_0 = \omega_e$	0.08	$=\omega_0$	$=\omega_0$	-794.31	

$H1: \omega_p \neq \omega_0 = \omega_e$	0.09	0.08	$=\omega_0$	-794.31	0
$H2: \omega_p = \omega_0 \neq \omega_e$	0.10	$=\omega_0$	0.06	-794.20	0.22
$H3: \omega_p \neq \omega_0 \neq \omega_e$	0.13	0.08	0.06	-794.10	0.42
OBP31 (within the genomic background chr 9)					
$H0: \omega_p = \omega_0 = \omega_e$	0.09	$=\omega_0$	$=\omega_0$	-1509.86	
$H1: \omega_p \neq \omega_0 = \omega_e$	0.08	0.10	$=\omega_0$	-1509.35	1.02
$H2: \omega_p = \omega_0 \neq \omega_e$	0.09	$=\omega_0$	0.08	-1509.82	0.08
$H3: \omega_p \neq \omega_0 \neq \omega_e$	0.06	0.10	0.08	-1509.19	1.34
OR36 (within the genomic background chr 5)					
$H0: \omega_p = \omega_0 = \omega_e$	0.41	$=\omega_0$	$=\omega_0$	-4737.69	
$H1: \omega_p \neq \omega_0 = \omega_e$	0.40	0.44	$=\omega_0$	-4737.57	0.24
$H2: \omega_p = \omega_0 \neq \omega_e$	0.40	$=\omega_0$	0.43	-4737.61	0.16
$H3: \omega_p \neq \omega_0 \neq \omega_e$	0.31	0.43	0.43	-4736.96	1.46

Note: chr – short of chromosome; *, ** - indicate the levels of significance of 0.05 and 0.01 respectively. ω_0 – dN/dS ratio for background branches; ω_p – dN/dS ratio for all *H. pardalinus* as a foreground branch; ω_e – dN/dS ratio for all *H. elevatus* as a foreground branches. P-value is calculated using likelihood ratio test (LRT) with the d.f.=1 for the comparisons of H0 vs. H1, H0 vs. H2, and d.f.=2 for the comparisons of H0 vs. H3.

Table S8. Test for positive selection using a branch-site model in PAML

	K0	K1	K2a	K2b	Sites ($\omega > 1$)	H0: $\ln L$	H1: $\ln L$	$2\Delta L$ (χ^2)
<i>OBP20</i>								
proportion _p	0.807	0.187	0.013	0.003	7 H** 111 N**	-797.38	-783.10	28.56**
foreground ω_p	0.0000	1.0000	115.56	115.56				
proportion _e	0.793	0.185	0.018	0.004	54 A**	-797.38	-791.36	12.04**
foreground ω_e	0.0000	1.0000	79.88	79.88				
<i>fzl</i>								
proportion _p	1.0000	0.0000	0.0000	0.0000		-1569.07	-1569.07	0
foreground ω_p	0.0000	1.0000	1.0000	1.0000				
proportion _e	0.7480	0.1701	0.0667	0.0152	33 P** 115 C* 138 N* 255 S* 388 S*	-2833.97	-2829.83	8.28**
foreground ω_e	0.0000	1.0000	7.55	7.55				

Note: Subscripts e and p indicate that a parameter was estimated for *H. elevatus* and *H. pardalinus* respectively; *, ** -point the level of significance of 0.05 and 0.01 respectively. The branch-site model uses four classes of sites: K0 - both foreground and background branch has sites that are under purifying selection ($\omega < 1$); K1 – sites of both foreground and background branches are under neutral selection ($\omega = 1$); K2a – sites on the foreground branch are under positive selection ($\omega > 1$) while sites on the background branch are under purifying selection ($\omega < 1$). K2b – sites on the foreground branch are under positive selection ($\omega > 1$) while sites on the background branch are under neutral selection ($\omega = 1$). H0 assumes that foreground branches may have different proportions of sites under neutral selection (K2a and K2b have $\omega=1$) to the background, while H1 assumes that the foreground branches may have a proportion of sites under positive selection (K2a and K2b have $\omega>1$). K0, K1, K2a, K2b are shown for H1 only. The comparisons of H0 vs. H1 is performed with d.f.=1 using LRT. The level of significance was calculated with χ^2 critical values of 3.84 at 5% and 5.99 at 1%. Bayes Empirical Bayes (BEB) analysis was used to single out sites under positive selection.

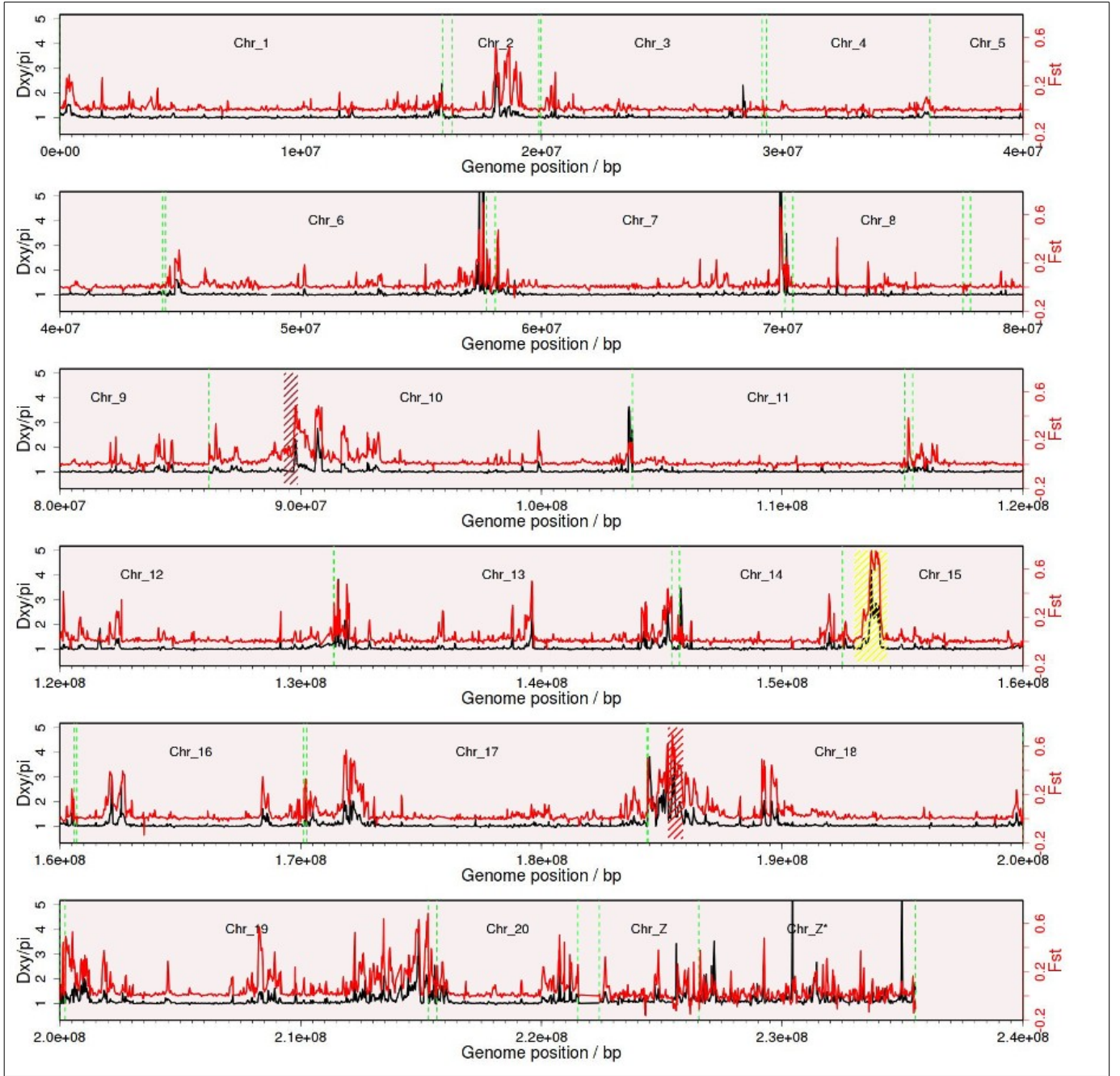


Figure S4. Genome scans from 50 kb moving windows showing F_{ST} (red graph) and D_{xy}/π (black graph). Both D_{xy}/π and F_{ST} represent the divergence between Amazonian *H. elevatus* and *H. pardalinus*. D_{xy} was corrected for π in *H. elevatus*. Shaded regions represent the Ac, N/Yb, and B/D colour pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and names by short name «Chr_*». The unmapped scaffolds of every chromosome are situated between the dash line at the end of every chromosome.

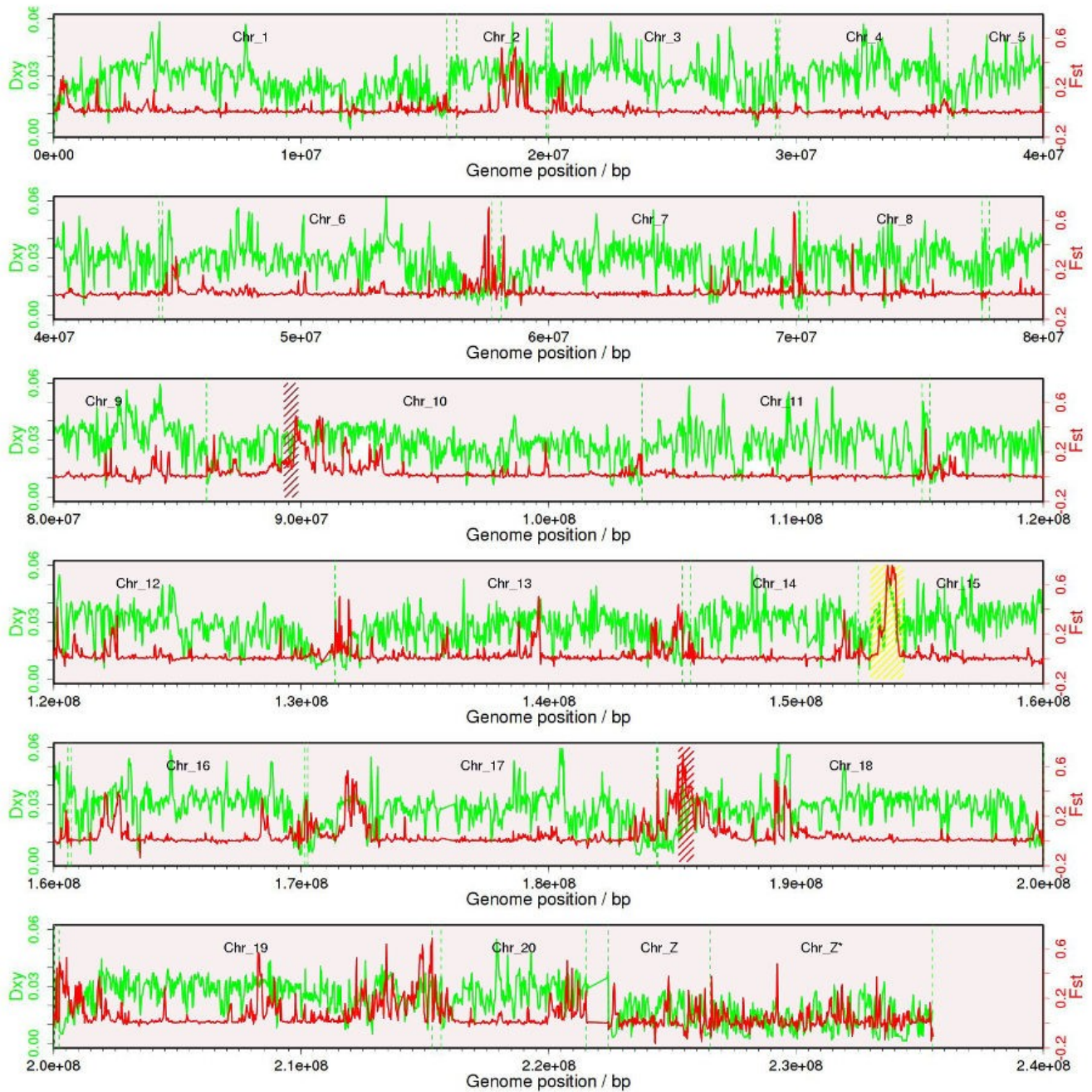


Figure S5. Genome scans from 50 kb moving windows showing F_{ST} (red graph) and D_{xy} (green graph). Both D_{xy} and F_{ST} represent the divergence between Amazonian *H. elevatus* and *H. pardalinus*. Shaded regions represent the *Ac*, *N/Yb*, and *B/D* colour pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and names by short name «Chr_*». The unmapped scaffolds of every chromosome are situated between the dash line at the end of every chromosome.

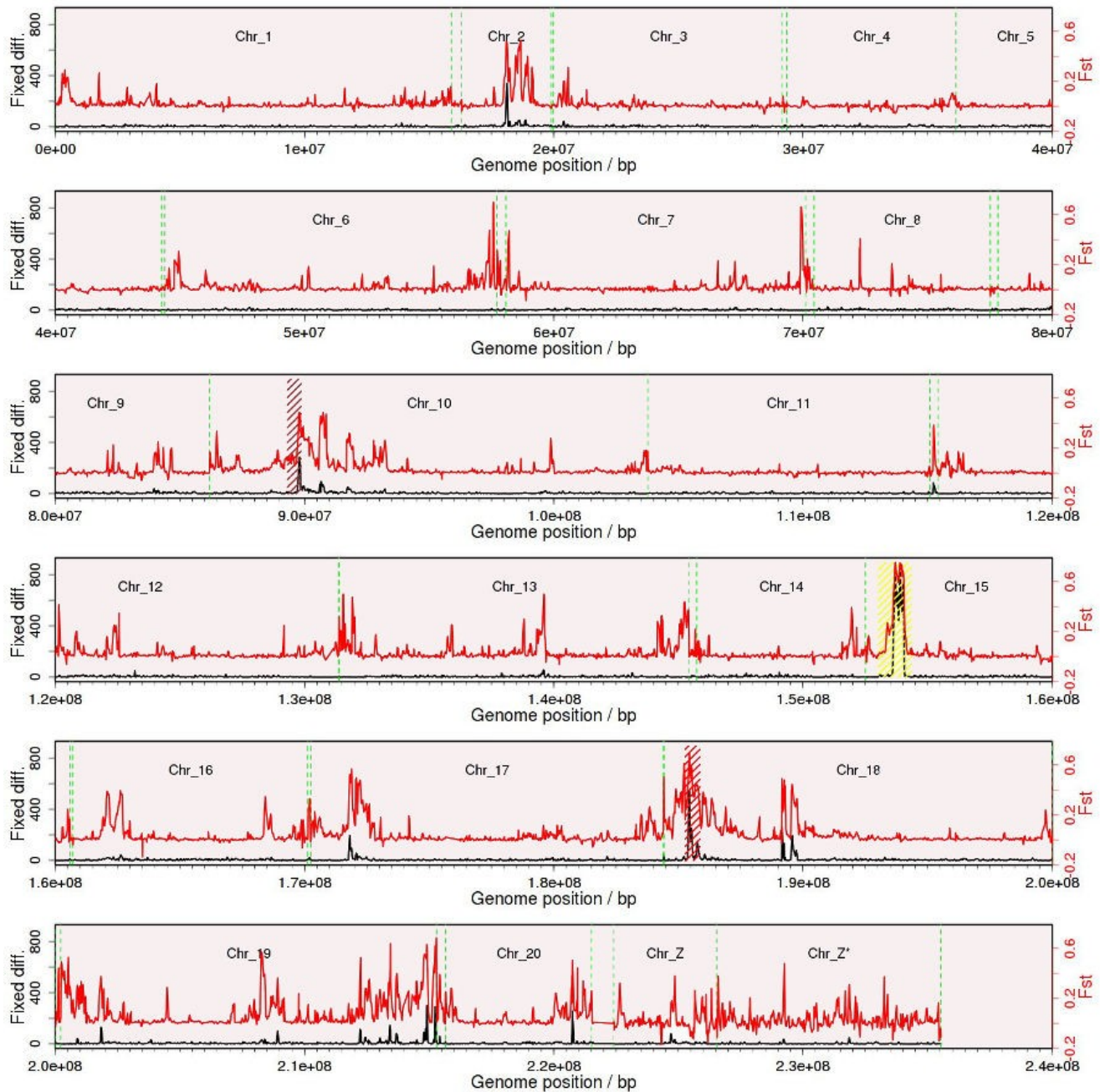


Figure S6. Genome scans from 50 kb moving windows showing F_{ST} (red graph) and number of fixed differences (black graph). Both F_{ST} and fixed differences represent the divergence between Amazonian *H. elevatus* and *H. pardalinus*. Shaded regions represent the Ac, N/Yb, and B/D colour pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and names by short name «Chr_*». The unmapped scaffolds of every chromosome are situated between the dash line at the end of every chromosome.

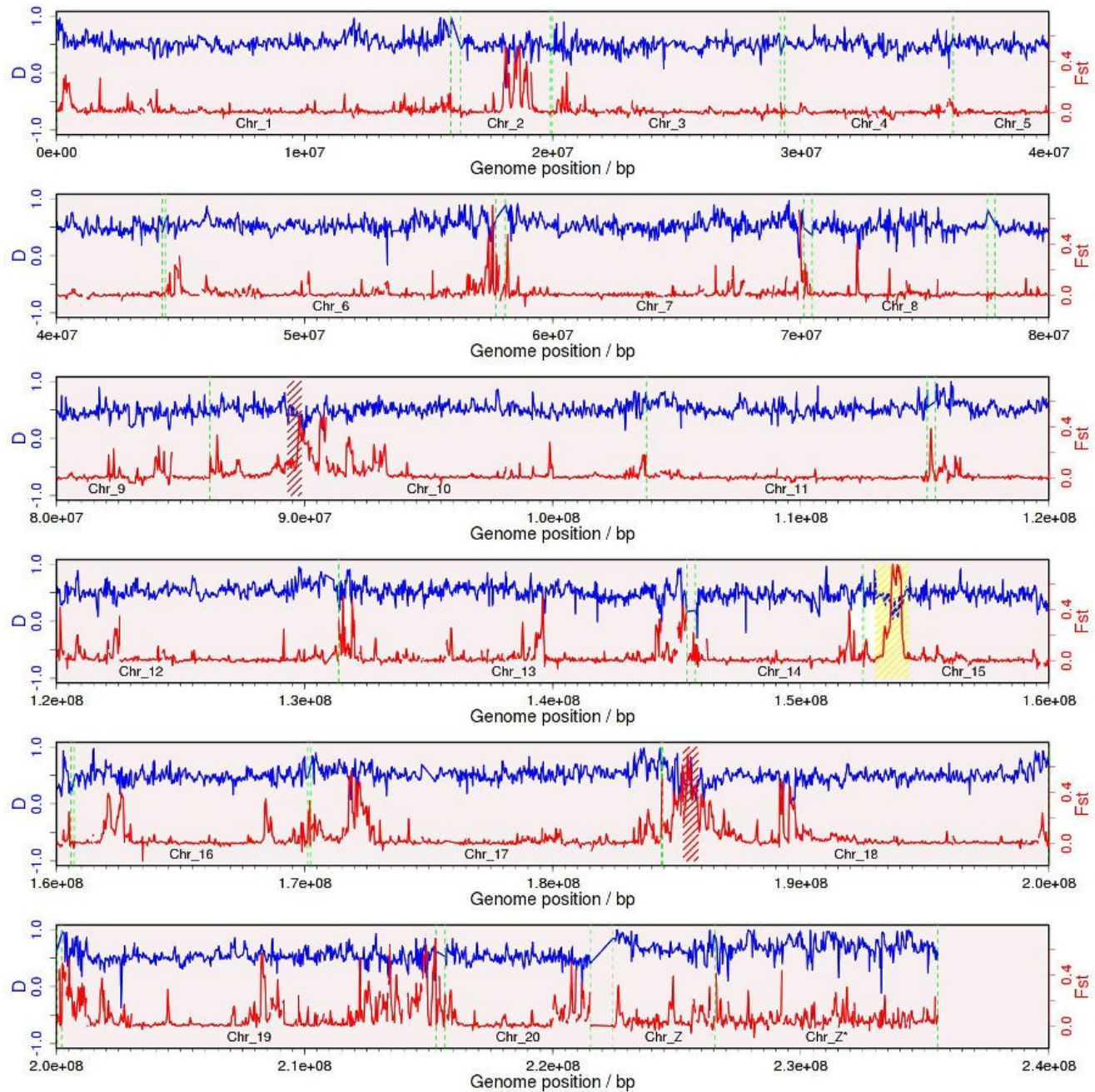


Figure S7. Distribution of F_{ST} (red graph) and D-statistics (blue graph) across the genome. The F_{ST} were derived from the comparison of Amazonian *H. elevatus* and *H. pardalinus*. D-statistics was calculated based on the tree ((Amazonian *H. pardalinus*), *H. p. sergestus*), (*H. elevatus bari*, (Amazonian *H. elevatus*))). Shaded regions represent the Ac, N/Yb, and B/D color pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and names by short name «Chr_*». The unmapped scaffolds of every chromosome are situated between the dash lines at the end of every chromosome. D-statistics vary around 0.5 along the whole genome indicating rampant admixture between Amazonian *H. elevatus* and *H. pardalinus*. F_{ST} and D-statistics demonstrate some degree of correlation when hollows of D coincide with peaks of F_{ST} .

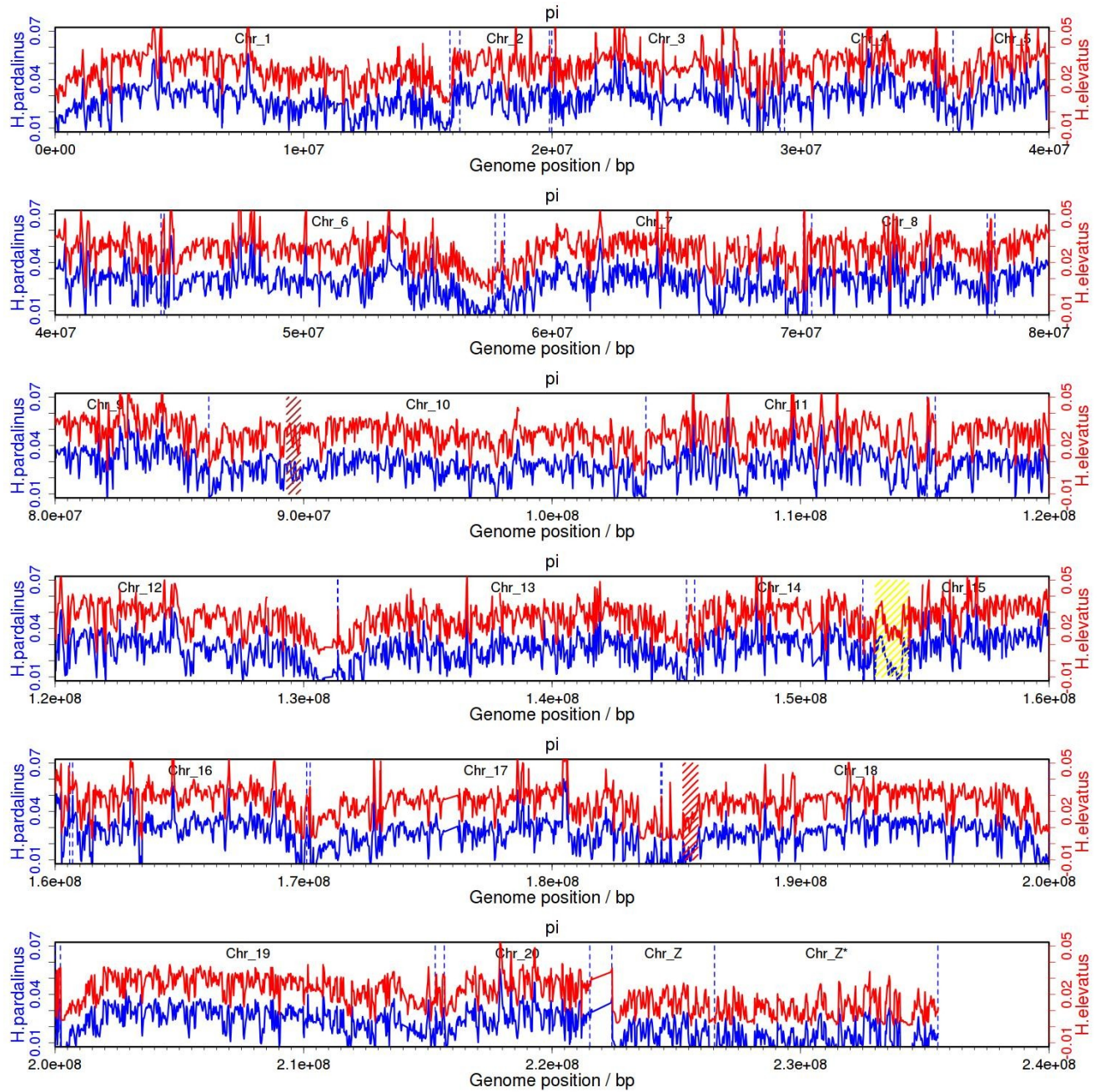


Figure S8. Genome scans from 50 kb moving windows showing π for *H. elevatus* (red graph) and *H. pardalinus* (blue graph). Shaded regions represent the Ac, N/Yb, and B/D colour pattern loci, on chromosomes 10, 15, and 18, respectively. The chromosomes are marked by dash line and names by short name «Chr_*». The unmapped scaffolds of every chromosome are situated between the dash lines at the end of every chromosome.

Fst peaks annotated

Chr	attributes	scaffold	start	end	Chrom_Pos	Genome_Pos	Description
chr2	HMEL015508	HE672041	29311	29381	1819170	18106001	
chr2	HMEL022562	HE672041	34334	36381	1825182	18112013	UDP-glycosyltransferase 33S3
chr2	HMEL015510	HE672041	44979	63358	1843993	18130824	
chr2	HMEL015511	HE672041	54781	54851	1844640	18131471	
chr2	HMEL015542	HE672041	378816	386049	2172257	18459088	
chr2	HMEL015543	HE672041	400905	401537	2191045	18477876	
chr2	HMEL015544	HE672041	401703	407873	2194612	18481443	
chr2	HMEL015545	HE672041	406382	410097	2198064	18484895	
chr2	HMEL017793	HE671660	51803	51875	2315063	18601894	
chr2	HMEL017792	HE671660	45449	101605	2336751	18623582	
chr2	HMEL017794	HE671660	103813	106439	2368350	18655181	
chr2	HMEL017795	HE671660	106851	110968	2372134	18658965	
chr2	HMEL008324	HE670771	265878	272336	2654680	18941511	
chr2	HMEL008325	HE670771	272566	277877	2660795	18947626	
chr2	HMEL008326	HE670771	284951	288670	2672384	18959215	
chr2	HMEL008327	HE670771	292841	298420	2681204	18968035	
chr6	HMEL014965	HE671768	115	479	12989568	57377329	
chr6	HMEL014966	HE671768	15273	44213	13019014	57406775	
chr6	HMEL004609	HE669659	37	3188	13155022	57542783	
chr6	HMEL004610	HE669659	26820	35489	13184564	57572325	
chr6	HMEL004611	HE669659	36402	37887	13190554	57578315	
chr6	HMEL004612	HE669659	42959	44638	13197208	57584969	
chr7	HMEL007657	HE670612	80329	81231	80780	58164120	
chr7	HMEL007658	HE670612	94369	104879	99624	58182964	
chr7	HMEL007659	HE670612	110017	114005	112011	58195351	
chr7	HMEL005299	HE670000	8603	9511	11817767	69901107	
chr7	HMEL005300	HE670000	11120	12406	11820473	69903813	
chr7	HMEL005301	HE670000	13057	16209	11823343	69906683	
chr7	HMEL005302	HE670000	16691	21996	11828054	69911394	
chr7	HMEL005303	HE670000	24241	32925	11837293	69920633	
chr7	HMEL005304	HE670000	38740	38810	11847485	69930825	
chr7	HMEL005305	HE670000	40658	59135	11858607	69941947	
chr7	HMEL005306	HE670000	70938	86784	11887571	69970911	
chr7	HMEL005307	HE670000	78990	79061	11887736	69971076	
chr7	HMEL005308	HE670000	89259	109441	11908060	69991400	
chr7	HMEL005309	HE670000	109582	113493	11920248	70003588	
chr7	HMEL005310	HE670000	113333	119579	11925166	70008506	
chr10	HMEL018100	HE668478	450400	483854	3608104	89789164	WntA
chr10	HMEL002910	HE668478	486270	489436	3628830	89809890	Chitin deacetylase homolog;
chr10	HMEL018101	HE668478	490610	494286	3633425	89814485	
chr10	HMEL018102	HE668478	495897	498544	3638198	89819258	
chr10	HMEL022560	HE671252	76116	78408	4457794	90638854	UDP-glycosyltransferase 33W1
chr10	HMEL012040	HE671252	82539	82609	4463106	90644166	
chr10	HMEL007028	HE670517	32134	35378	4499311	90680371	
chr10	HMEL007029	HE670517	43579	43653	4509171	90690231	
chr10	HMEL007030	HE670517	50255	50325	4515845	90696905	
chr10	HMEL022672	HE670517	50986	53212	4517654	90698714	UDP-glycosyltransferase 33U1
chr10	HMEL022655	HE670517	54461	57956	4521764	90702824	UDP-glycosyltransferase 33U2
chr10	HMEL022629	HE670517	59090	61651	4525926	90706986	UDP-glycosyltransferase 33U3
chr10	HMEL022644	HE670517	63760	66302	4530586	90711646	UDP-glycosyltransferase 33U4
chr10	HMEL022633	HE670517	73534	77119	4540882	90721942	UDP-glycosyltransferase 33AB3
chr10	HMEL007032	HE670517	90377	90447	4555967	90737027	
chr10	HMEL022620	HE670517	92269	94685	4559032	90740092	UDP-glycosyltransferase 33AA1
chr10	HMEL022626	HE670517	102058	104507	4568838	90749898	UDP-glycosyltransferase 33Y1
chr10	HMEL022631	HE670517	105661	107568	4572170	90753230	UDP-glycosyltransferase 33Y2
chr10	HMEL022649	HE670517	109803	114398	4577656	90758716	UDP-glycosyltransferase 33V1
chr10	HMEL022642	HE670517	124431	127213	4591377	90772437	UDP-glycosyltransferase 33V2
chr10	HMEL007035	HE670517	138999	141599	4605854	90786914	
chr10	HMEL022614	HE670517	213709	216512	4680666	90861726	UDP-glycosyltransferase 33U5
chr10	HMEL007040	HE670517	216032	216105	4681624	90862684	
chr12	HMEL008576	HE670801	23319	23389	4722990	120142785	

Fst peaks annotated

chr12	HMEL008577	HE670801	47693	47763	4747364	120167159	
chr13	HMEL017509	HE671624	213558	213628	8208146	139592373	
chr13	HMEL017508	HE671624	207746	222286	8209569	139593796	
chr13	HMEL017510	HE671624	223227	223297	8217815	139602042	
chr13	HMEL017511	HE671624	228596	233081	8225392	139609619	
chr13	HMEL010567	HE671097	128589	131468	13828996	145213223	
chr13	HMEL010568	HE671097	133950	157667	13844776	145229003	
chr13	HMEL010569	HE671097	154458	154528	13853460	145237687	
chr13	HMEL010570	HE671097	182978	185467	13883190	145267417	
chr13	HMEL010571	HE671097	195326	197089	13895175	145279402	
chr13	HMEL010572	HE671097	197152	197901	13896494	145280721	
chr15	HMEL000022	HE667780	605562	611917	1117312	153624011	putative enoyl-CoA hydratase
chr15	HMEL000023	HE667780	612159	613439	1121371	153628070	
chr15	HMEL000024	HE667780	613478	616174	1123398	153630097	similar to: Sur-8
chr15	HMEL002023	HE667780	619896	623177	1130109	153636808	
chr15	HMEL002024	HE667780	624527	627276	1134474	153641173	
chr15	HMEL002025	HE667780	668996	669066	1177603	153684302	
chr15	HMEL000025	HE667780	670615	681472	1184616	153691315	fizzy-like
chr15	HMEL002026	HE667780	788769	788839	1297376	153804075	
chr15	HMEL002027	HE667780	804540	804610	1313147	153819846	
chr15	HMEL002028	HE667780	841353	841465	1349981	153856680	
chr15	HMEL002029	HE667780	843550	843630	1352162	153858861	
chr15	HMEL002030	HE667780	843861	844085	1352545	153859244	
chr15	HMEL000026	HE667780	843832	867595	1364286	153870985	putative poly(A)-specific ribonuclease
chr15	HMEL000027	HE667780	868403	874957	1380252	153886951	
chr15	HMEL000028	HE667780	875179	876242	1384283	153890982	similar to: ARP-like
chr15	HMEL000029	HE667780	876837	877934	1385958	153892657	
chr15	HMEL002036	HE667780	878290	879387	1387411	153894110	
chr15	HMEL000030	HE667780	878156	879586	1387443	153894142	putative proteasome 26Snon ATPase subunit 4
chr15	HMEL000031	HE667780	879392	886137	1391337	153898036	
chr15	HMEL000032	HE667780	887765	898284	1401597	153908296	putative zinc phosphodiesterase
chr15	HMEL000033	HE667780	898556	947644	1431672	153938371	similar to: serine/threonine-protein kinase
chr15	HMEL000034	HE667780	949155	958304	1462302	153969001	WD repeat domain 13 containing
chr15	HMEL000036	HE667780	969404	995419	1490984	153997683	similar to: WAS protein family homologue 1
chr15	HMEL000037	HE667780	992698	999815	1504829	154011528	putative Tyrosine phosphatase
chr15	HMEL000038	HE667780	1000333	1020667	1519072	154025771	similar to: lethal (2) k05819
chr15	HMEL000039	HE667780	1020412	1023980	1530768	154037467	putative mitogen-activated protein kinase
chr15	HMEL000040	HE667780	1024491	1027417	1534526	154041225	putative DNA excision repair protein
chr15	HMEL000041	HE667780	1028128	1032655	1538964	154045663	similar to: penguin
chr15	HMEL000042	HE667780	1032646	1036017	1542904	154049603	putative thymidylate kinase
chr15	HMEL000043	HE667780	1036206	1038564	1545957	154052656	putative caspase-activated DNase
chr15	HMEL000044	HE667780	1043068	1045547	1552880	154059579	
chr15	HMEL000045	HE667780	1045873	1048387	1555702	154062401	
chr15	HMEL000046	HE667780	1046944	1048299	1556194	154062893	
chr17	HMEL014316	HE671441	8306	8375	1531194	171794367	
chr17	HMEL014317	HE671441	81141	81210	1604029	171867202	
chr17	HMEL014318	HE671441	128441	128512	1651330	171914503	
chr17	HMEL008286	HE670767	36966	51752	1814466	172077639	
chr17	HMEL013014	HE671321	38984	64123	1921011	172184184	
chr17	HMEL013016	HE671321	85665	85735	1955157	172218330	
chr17	HMEL013017	HE671321	92950	93020	1962442	172225615	
chr17	HMEL013015	HE671321	67875	122466	1964628	172227801	
chr17	HMEL013018	HE671321	99797	99867	1969289	172232462	
chr17	HMEL013019	HE671321	100891	100961	1970383	172233556	
chr18	HMEL013552	HE671969	171448	172906	476686	184899976	
chr18	HMEL013553	HE671969	181816	197841	494338	184917628	
chr18	HMEL003859	HE669320	70057	70180	775446	185198736	
chr18	HMEL003860	HE669320	74913	85445	785506	185208796	
chr18	HMEL003861	HE669320	86040	89259	792977	185216267	
chr18	HMEL003862	HE669320	91314	94471	798220	185221510	
chr18	HMEL003863	HE669320	99372	137837	823932	185247222	
chr18	HMEL001033	HE670865	515905	518516	929223	185352513	
chr18	HMEL001034	HE670865	513632	518516	930359	185353649	

Fst peaks annotated

chr18	HMEL001034	HE670865	513632	518516	930359	185353649	
chr18	HMEL001035	HE670865	512184	513156	933763	185357053	
chr18	HMEL001036	HE670865	508486	509374	937503	185360793	
chr18	HMEL001037	HE670865	501570	508263	941517	185364807	
chr18	HMEL001038	HE670865	495971	501266	947815	185371105	
chr18	HMEL001039	HE670865	487907	495751	954604	185377894	
chr18	HMEL001042	HE670865	481751	486547	962284	185385574	
chr18	HMEL001043	HE670865	478981	481565	966160	185389450	
chr18	HMEL001044	HE670865	467864	478657	973173	185396463	
chr18	HMEL001028	HE670865	438423	439107	1007668	185430958	optix
chr18	HMEL002051	HE670865	413693	413765	1032704	185455994	
chr18	HMEL002050	HE670865	409998	410068	1036400	185459690	
chr18	HMEL002049	HE670865	373423	373493	1072975	185496265	
chr18	HMEL002048	HE670865	359640	359710	1086758	185510048	
chr18	HMEL002047	HE670865	358751	358821	1087647	185510937	
chr18	HMEL001014	HE670865	306696	311266	1137452	185560742	
chr18	HMEL001006	HE670865	202065	203349	1243726	185667016	
chr18	HMEL001004	HE670865	196468	200906	1247746	185671036	
chr18	HMEL002045	HE670865	173366	173434	1273033	185696323	
chr18	HMEL015738	HE671526	19285	19355	4729879	189153169	
chr18	HMEL015739	HE671526	22884	22953	4733478	189156768	
chr18	HMEL015740	HE671526	35731	37580	4747215	189170505	
chr18	HMEL003081	HE668655	4907	10542	4843532	189266822	
chr18	HMEL017512	HE671625	5858	8189	5096670	189519960	
chr18	HMEL017513	HE671625	64452	81700	5162722	189586012	
chr18	HMEL017514	HE671625	89435	89505	5179116	189602406	
chr19	HMEL008682	HE670819	1660	4804	3232	200215468	
chr19	HMEL008683	HE670819	7204	16974	12089	200224325	
chr19	HMEL008684	HE670819	25781	25852	25817	200238053	
chr19	HMEL008685	HE670819	34321	72594	53458	200265694	
chr19	HMEL008686	HE670819	72836	77534	75185	200287421	
chr19	HMEL008687	HE670819	78512	78582	78547	200290783	
chr19	HMEL008689	HE670819	84925	85329	85127	200297363	Odorant binding protein OBP20
chr19	HMEL003062	HE668621	35426	41260	145468	200357704	
chr19	HMEL003063	HE668621	46276	48750	154638	200366874	
chr19	HMEL012019	HE671250	139170	139889	300563	200512799	
chr19	HMEL012020	HE671250	141138	144344	303774	200516010	
chr19	HMEL017987	HE671693	1833	1904	8029949	208242185	
chr19	HMEL017988	HE671693	2064	2135	8030180	208242416	
chr19	HMEL017989	HE671693	2839	18955	8038977	208251213	
chr19	HMEL017989	HE671693	2839	18955	8038977	208251213	
chr19	HMEL017990	HE671693	45785	59772	8080859	208293095	
chr19	HMEL017991	HE671693	61206	66222	8091794	208304030	
chr19	HMEL017992	HE671693	81343	84842	8111173	208323409	
chr19	HMEL017993	HE671693	103002	103074	8131118	208343354	
chr19	HMEL008770	HE670829	276044	282180	12019168	212231404	Cytochrome p450 4CG4
chr19	HMEL008771	HE670829	284675	284746	12024767	212237003	
chr19	HMEL008772	HE670829	285361	286728	12026101	212238337	Cytochrome p450 4CG12
chr19	HMEL010694	HE671117	29406	31842	13217764	213430000	
chr19	HMEL010695	HE671117	33357	36525	13222081	213434317	
chr19	HMEL006026	HE670348	55957	88526	13494464	213706700	
chr19	HMEL006027	HE670348	89211	92482	13513069	213725305	Olfactory receptor OR30
chr19	HMEL006028	HE670348	93452	99058	13518477	213730713	
chr19	HMEL005296	HE669998	32657	32730	14556447	214768683	
chr19	HMEL005297	HE669998	69006	75704	14596108	214808344	
chr19	HMEL018008	HE671939	24668	24738	14688456	214900692	
chr19	HMEL018009	HE671939	55205	55275	14718993	214931229	
chr19	HMEL018010	HE671939	64188	66482	14729088	214941324	
chr19	HMEL013381	HE671345	56254	62042	14944973	215157209	
chr19	HMEL005561	HE670118	21481	21551	15027453	215239689	
chr19	HMEL005562	HE670118	33290	34486	15039825	215252061	
chr20	HMEL014563	HE671456	32135	33948	5087197	220748224	
chr20	HMEL014562	HE671456	27236	31956	5090642	220751669	
chr20	HMEL013073	HE671330	152203	154753	5273815	220934842	

Fst peaks and corresponding chromosome are marked by PEAK..CHR., respectively

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